

CS 6120

Natural Language Processing

Instructor: Amir Tahmasebi, PhD

Fall 2026



Course Logistics: *Organization*

- *Contact:*
 - a.tahmasebi@northeastern.edu
- *Class schedule:*
 - Tuesdays 6:00pm-9:20pm ET
- *Presentation*
 - *In person*
- *Office hours:*
 - *TAs: see next slide*
 - *Instructor:*
 - *every teaching session after the class end*
 - *1:1 per request on zoom*

Course Logistics: TA



Zankhana Pratik Mehta

mehta.zan@northeastern.edu



Craig Roberts

roberts.cr@northeastern.edu



TA responsibilities

- Office Hours
- Lab sessions/Tutorials
- Assignment markings
- [Final project markings]

Course Logistics: *Technical*



Announcements
Assignments
Quizzes
Submissions
Discussion

Course Logistics: *Grading*

- *Assignments (5: 40%)*
- *Pop Quizzes (5: 20%)*
- *Class attendance and participation (5%)*
- *Class Presentation (5%)*
- *Final Project (1: 30%)*

Homework Submissions

- **Posting and Access:** Canvas
- **Solution Submission:** Canvas
- **Solution Format:**

REPORT:

- .docx
- .pdf
- .txt
- .tex
- .zip/.tar

CODE:

- .py
- .ipynb
- .zip/.tar

Quizzes

Total Mark: 20%

Five Quizzes*

Multiple Choice

In Class

On Canvas

Requires access code that will be provided in the class

* Answers will NOT be shared.

Student Class Presentations

Total Mark: 5%

Topic: Publication will be assigned by Instructor

Format: 1 slide in class presentation (2mins)

Team Size: every student individually

Final Project

Total Mark: 30%

Format: in-class short presentation (last two sessions of the course) + Report + Code

Team size*: 2-3 students

Topics:

- Proposed by Instructor (will be shared mid semester)
- Proposed by Students

* No individual projects are accepted.

Class Participation

Total Mark: 5%

In Person Attendance

Random Presence Check on Canvas

Requires access code that will be provided in the class

Class Engagement

NOTE: Absences should be submitted on designated teams form (see next slide).

More than 4 absences will automatically result in failing the course.

Course Logistics: *Absence*

Absence Submission

- Fill out a form on teams (to be shared)
- Should be requested per session prior to the session start

Course Logistics: *Grading*

Grades

Grade	Count	Passing
A		✓
A-		✓
B		✓
B-		✓
B+		✓
C		✓
C-		✓
C+		✓
D		✗
D-		✗
D+		✗
F		✗
I		✗
IP		✗
NE		✗
S		✓
U		✗
W		✗
X		✗

93-100

90-92

87-89

83-86

80-82

77-79

73-76

70-72

67-69

63-66

60-62

< 59

A

A-

B+

B

B-

C+

C

C-

D+

D

D-

F

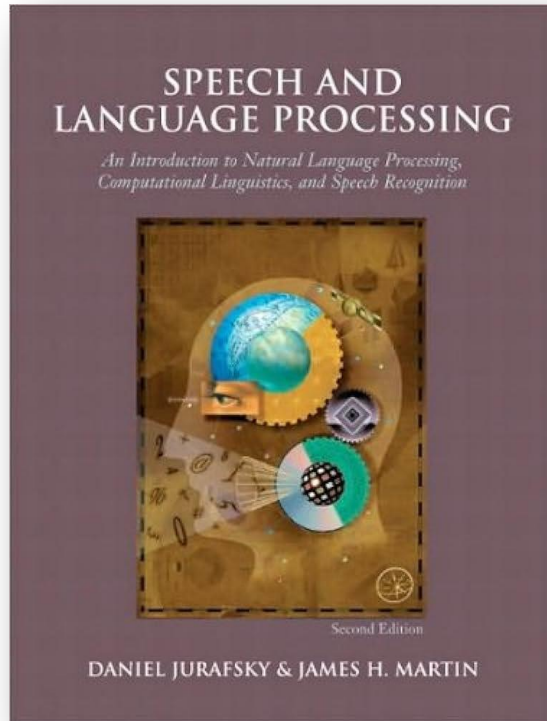
AI Tool Usage Policy

The Core Principle

You must be the intellectual owner of every line you submit.

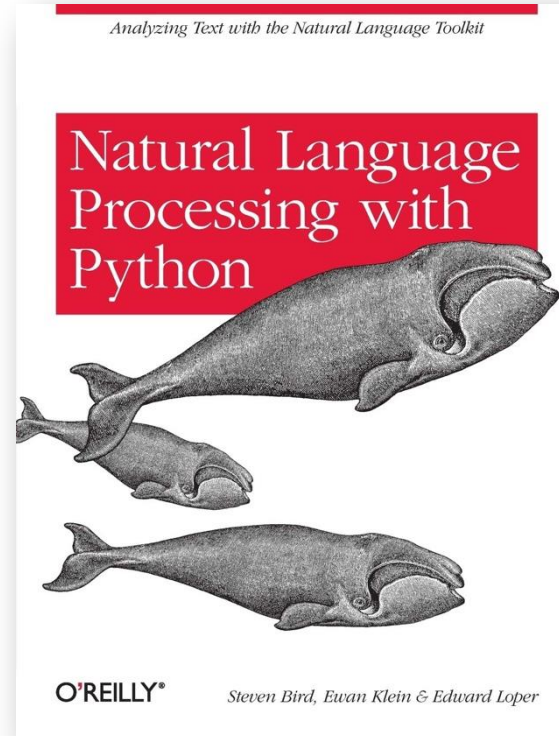
AI tools may assist your process, but the understanding, design decisions, and final code or writing must demonstrably be yours. If you cannot explain, modify, or defend any part of your submission in detail, it is considered a violation of this policy regardless of how it was produced.

References



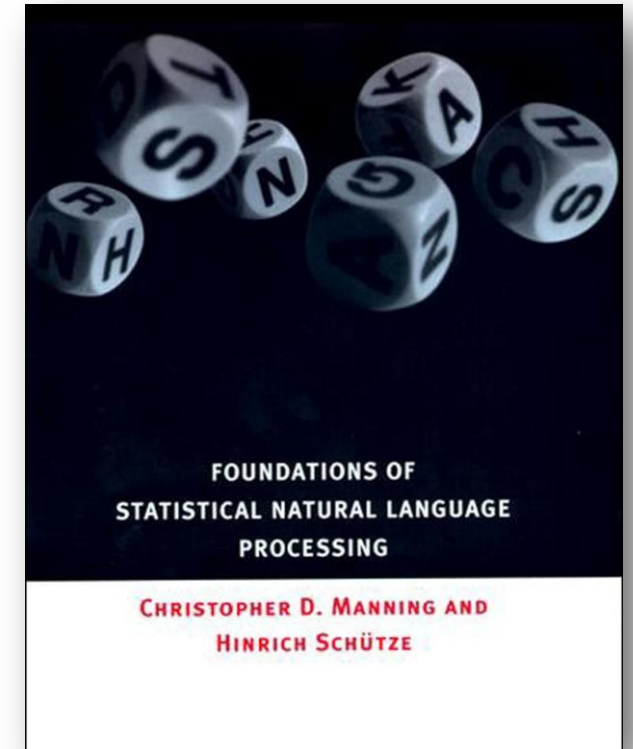
[3rd edition](#)

Speech and Language Processing
Dan Jurafsky and James H. Martin



[link](#)

Natural Language Processing
with Python" by Steven Bird et
al.



[link](#)

Foundations of Statistical Natural
Language Processing" by Christopher D.
Manning and Hinrich Schütze

References

- Publications in conferences like ACL (Association for Computational Linguistics), NAACL, EMNLP, NeurIPS
- Tutorials and documentation for NLP frameworks like TensorFlow, PyTorch, SpaCy, and Hugging Face's Transformers

Syllabus

- What is Natural Language Processing?
 - History
 - Applications
- Text Processing Fundamentals
 - Regular Expressions
 - Tokenization
 - Normalization
 - Libraries and Toolkits
- Syntax and Grammar
 - POS tagging
 - Syntax tree
- Word Representation
 - TF-IDF
 - word2vec
 - Glove
- Intro to Machine Learning
 - Logistic Regression (LR)
 - Neural Networks
 - Recurrent Neural Networks (RNN)
 - Convolutional Neural Networks (CNN)
 - Transformers
- Language Models
- NLP Use Cases
 - Named Entity Recognition
 - Machine Translation
 - QA
- Large Language Models

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Course Calendar

WEEK 1: Introduction to NLP

WEEK 2: Text Processing Fundamentals

HW1

WEEK 3: Pytorch Tutorial

(HW1 Due)

WEEK 4: Word Representation

HW2

WEEK 5: Machine Learning Primer (part I)

(HW2 Due) Final Project Topic Sharing

WEEK 6: Machine Learning Primer (part II)

HW4

Course Calendar

WEEK 7: Language Models

HW5, (HW4 Due)

WEEK 8: NLP Use Cases

(HW5 Due)

(Final Project Topic Selection Due)

WEEK 9: NLP Use Case + [Guest Speaker]

WEEK 10: NLP Use Case

WEEK 11: NLP Use Case + [Guest Speaker]

WEEK 12: Large Language Models

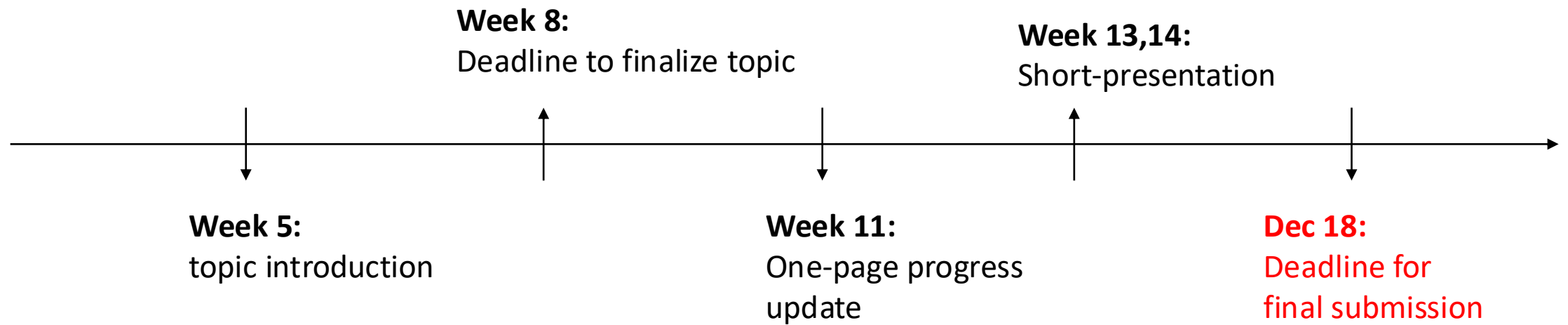
WEEK 13,14: Final Project Presentations

LAST DAY OF CLASS:

Tuesday Dec 8, 2026

Final Project

Timeline



Natural Language Processing

NLP to LLM and AI Agents

- LLMs are black boxes that fail silently

classical NLP gives you the tools to diagnose why



Natural Language Processing

NLP to LLM and AI Agents

- Does the output *LOOK* good?

Evaluation is classical NLP's home turf

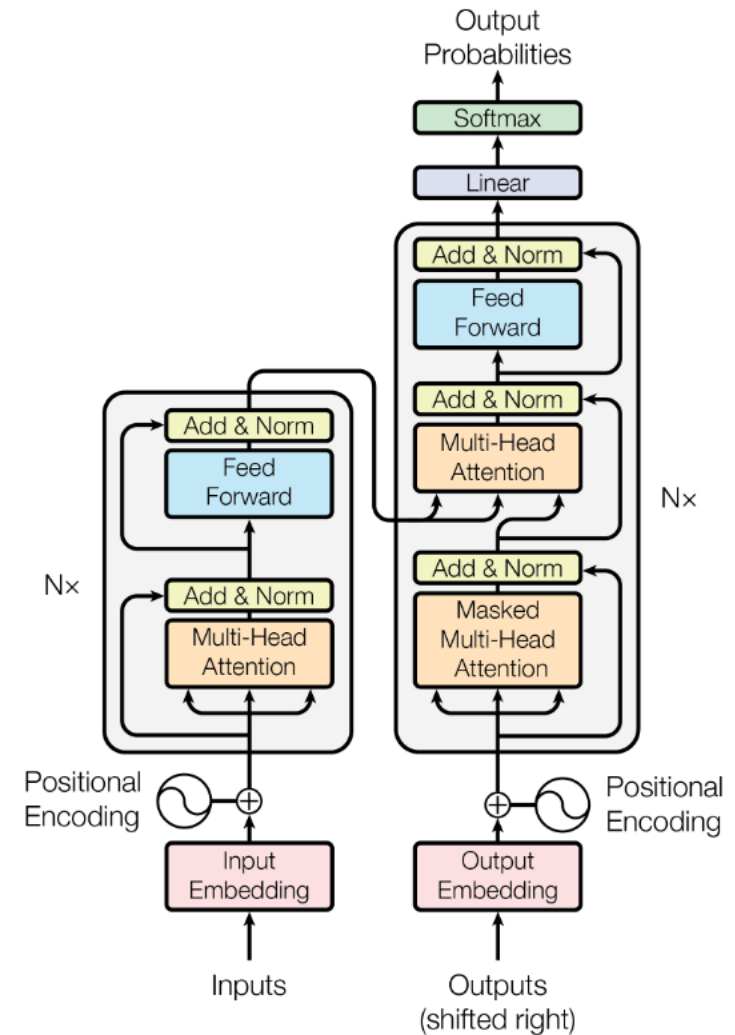


Natural Language Processing

NLP to LLM and AI Agents

- Distributional semantics → word2vec → transformers

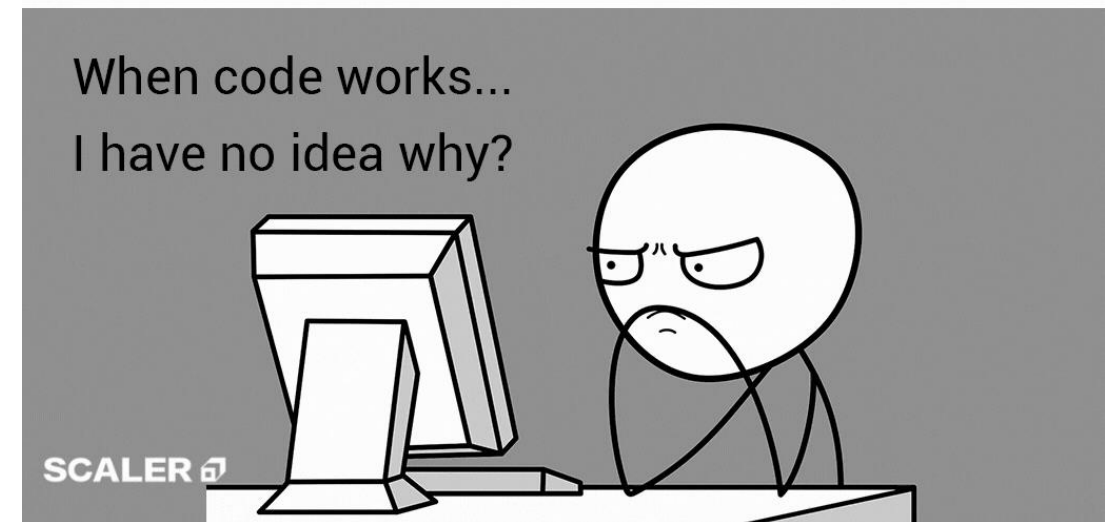
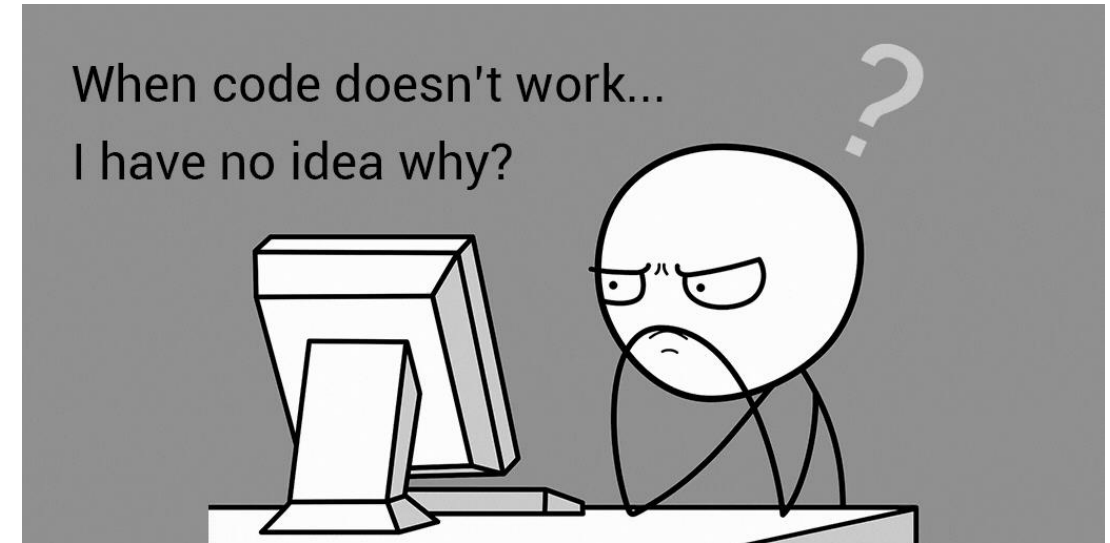
Understanding why embeddings work, and how the field moved from count-based to predictive to contextual representations, is what lets a student read a new paper in 2027 and place it, rather than treating each new model as a magic upgrade.



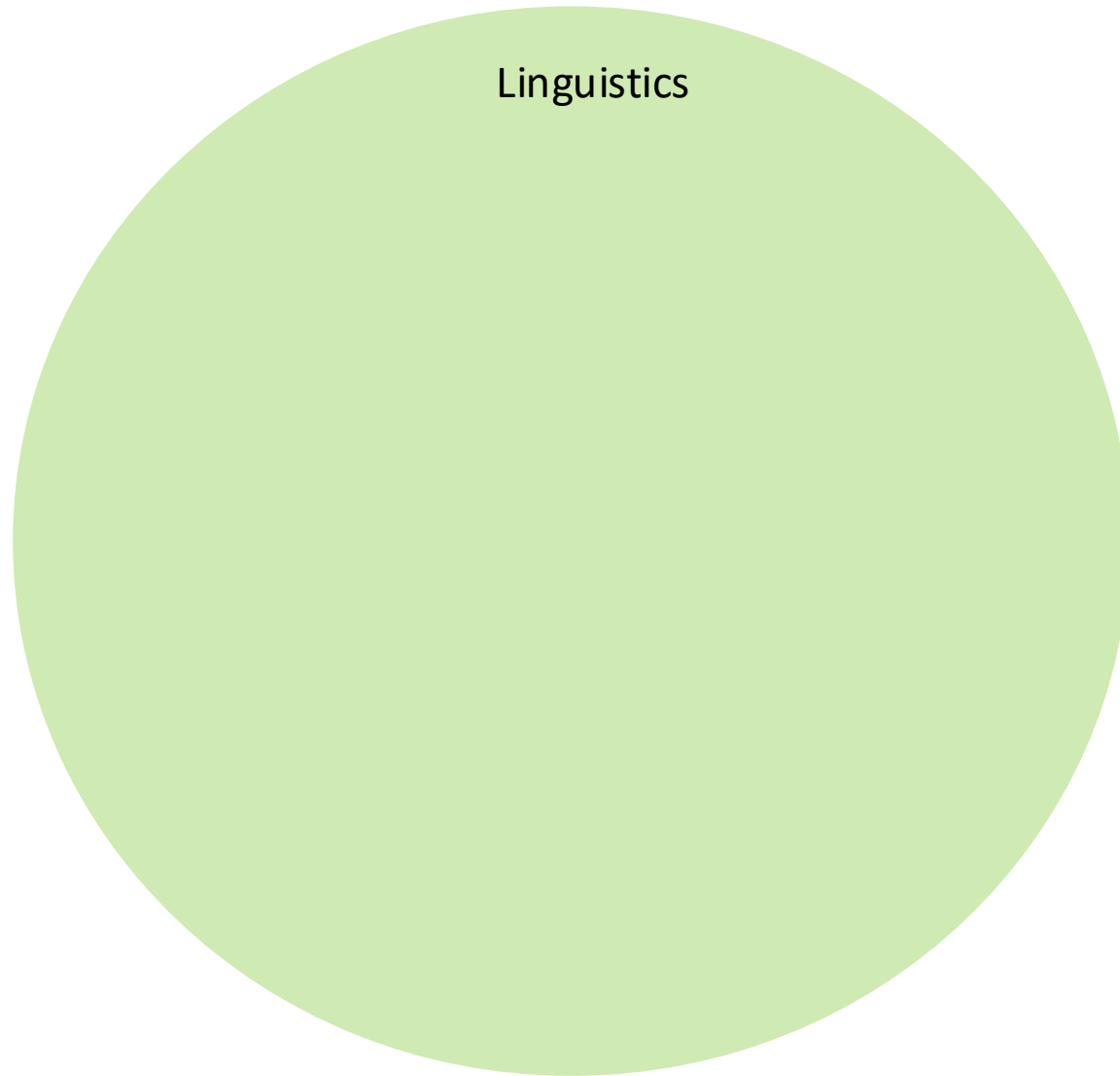
Natural Language Processing

NLP to LLM and AI Agents

- NOT as simple as an API call...

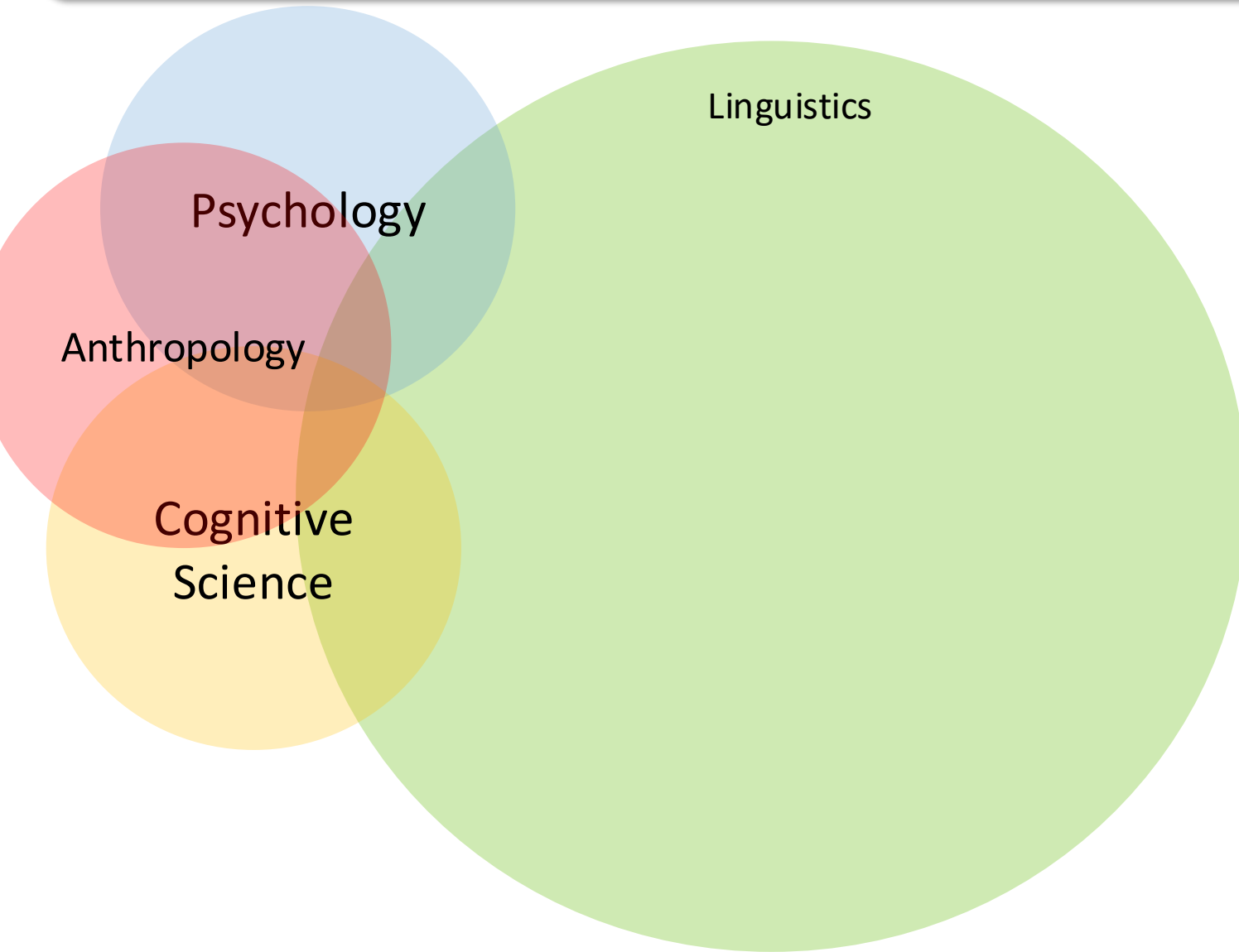


Natural Language Processing

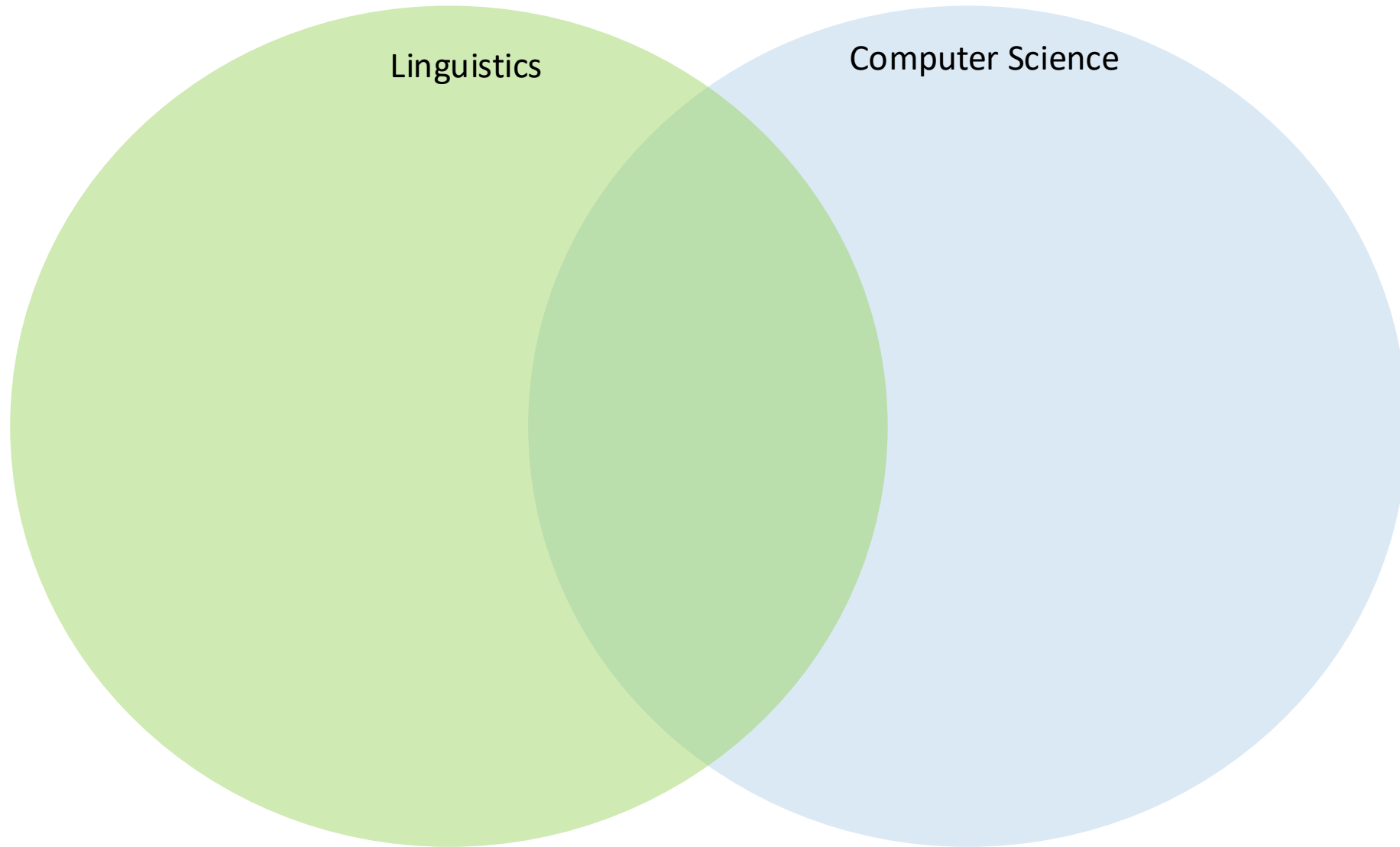


scientific study
of language

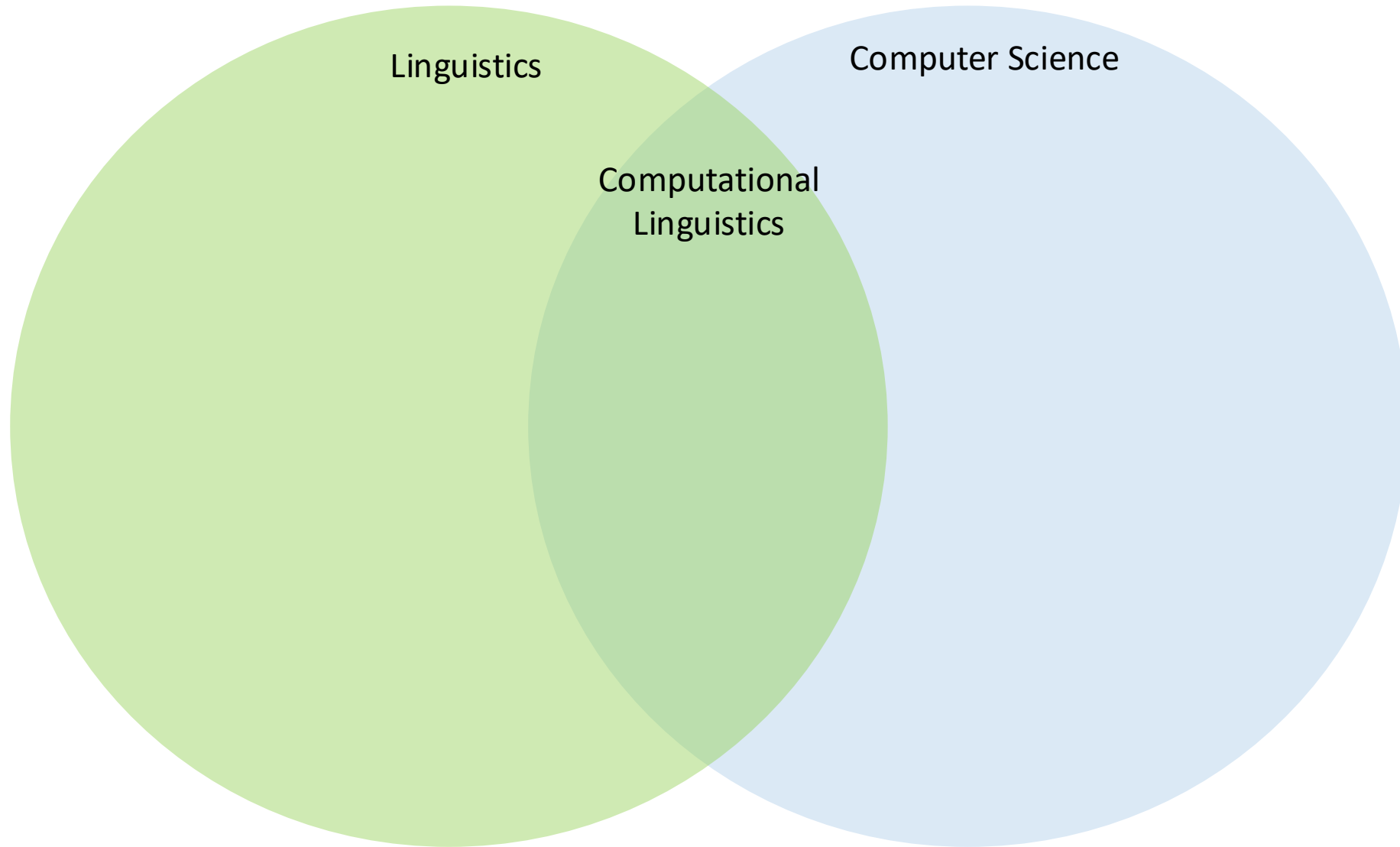
Natural Language Processing



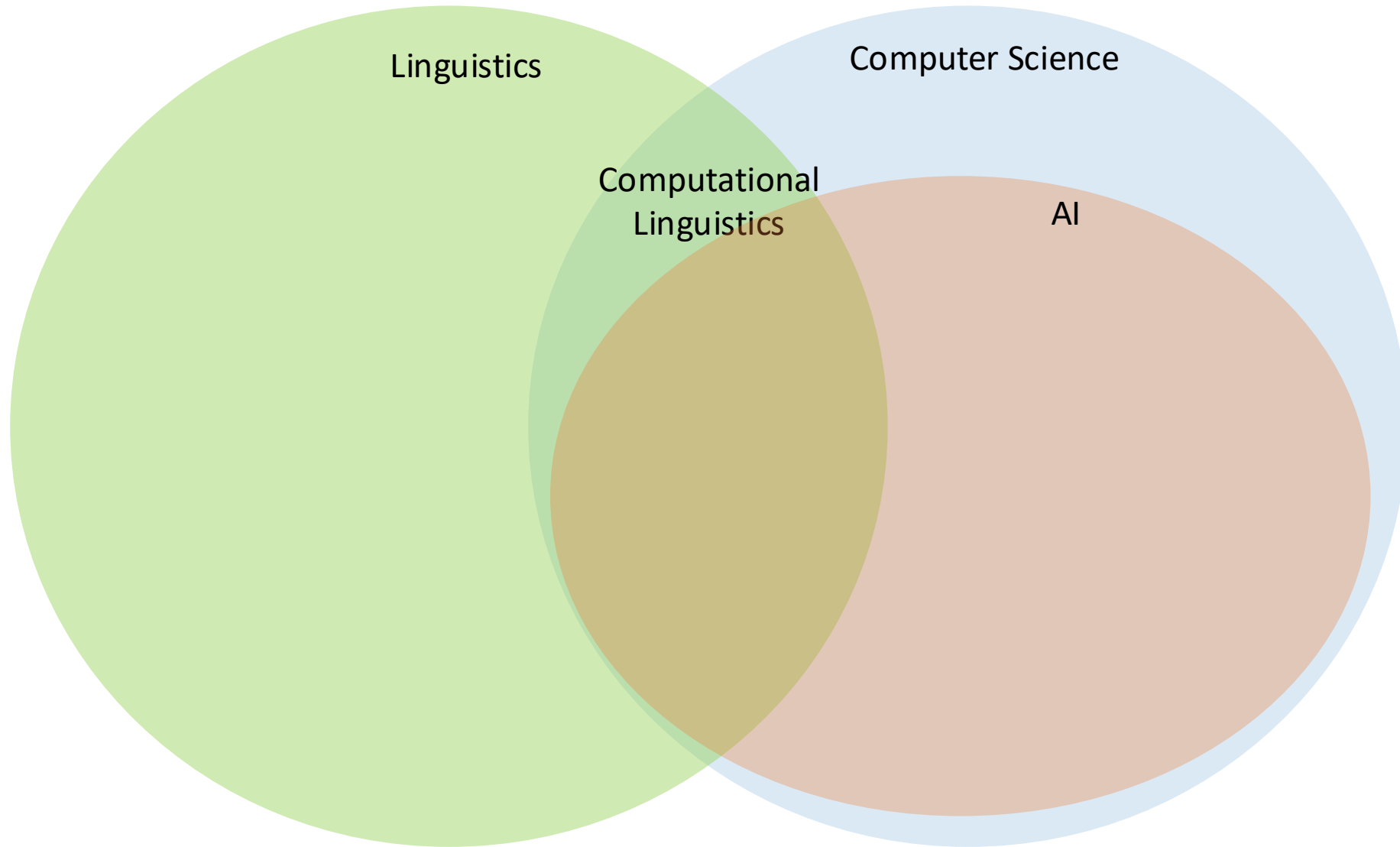
Natural Language Processing



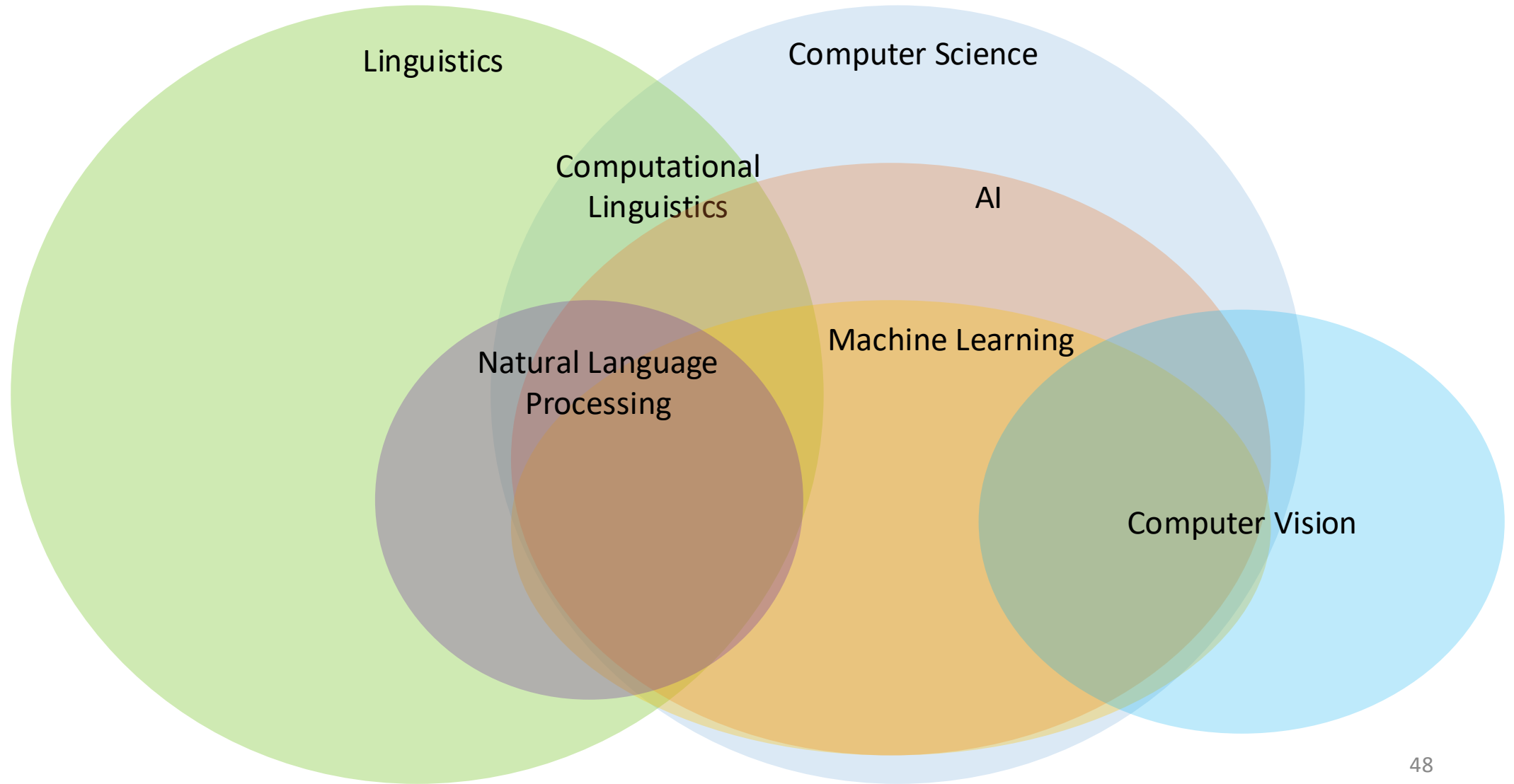
Natural Language Processing



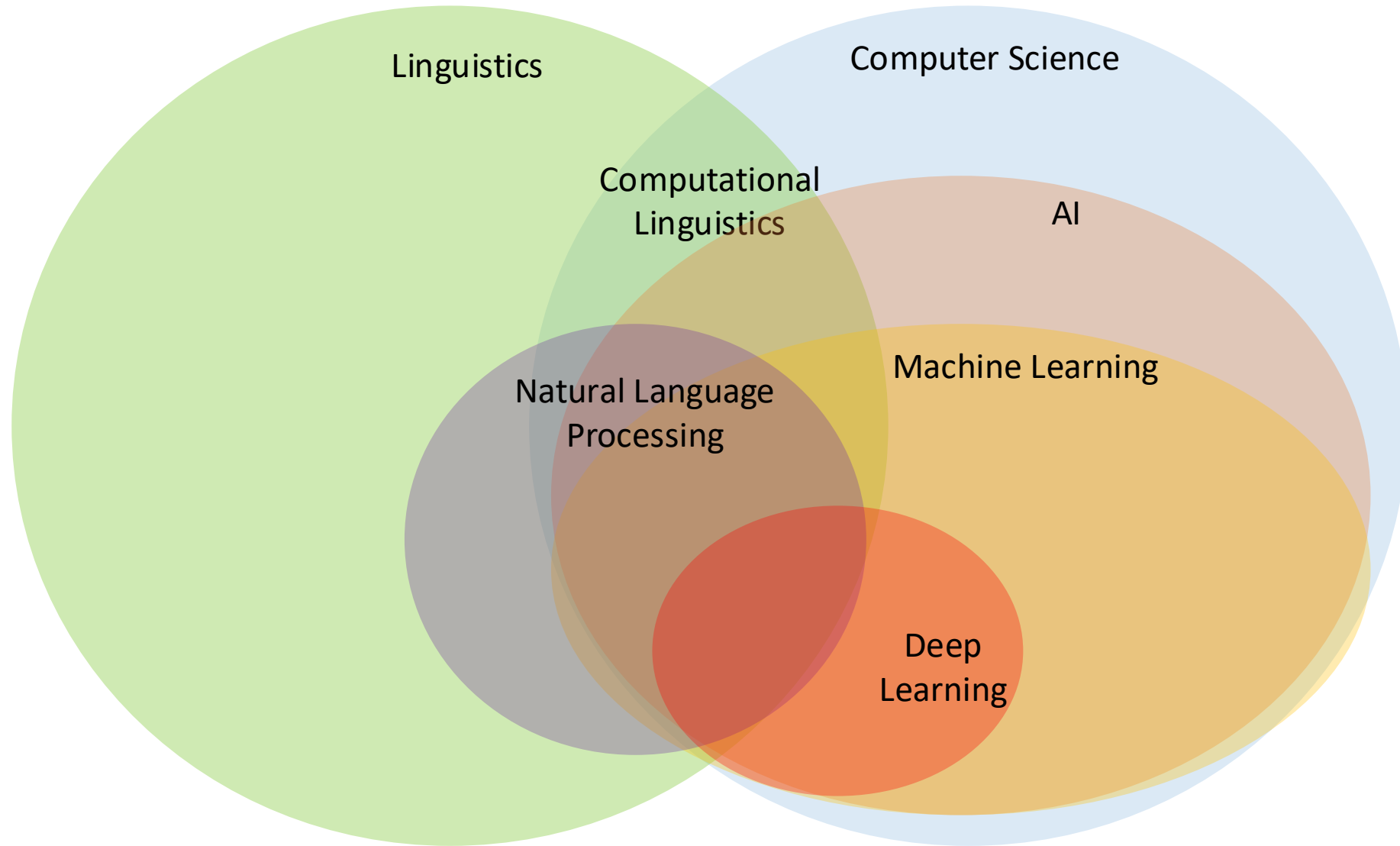
Natural Language Processing



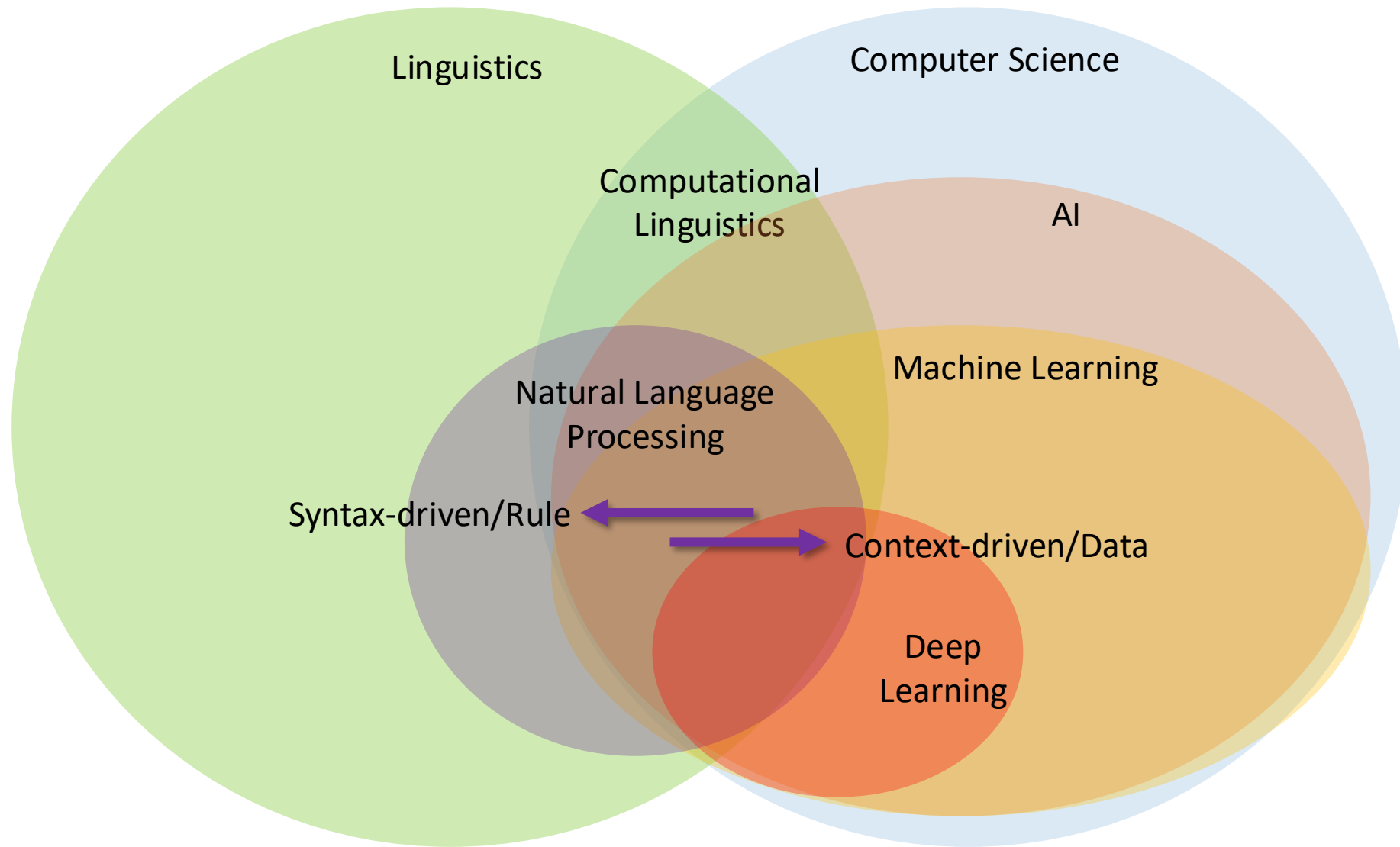
Natural Language Processing



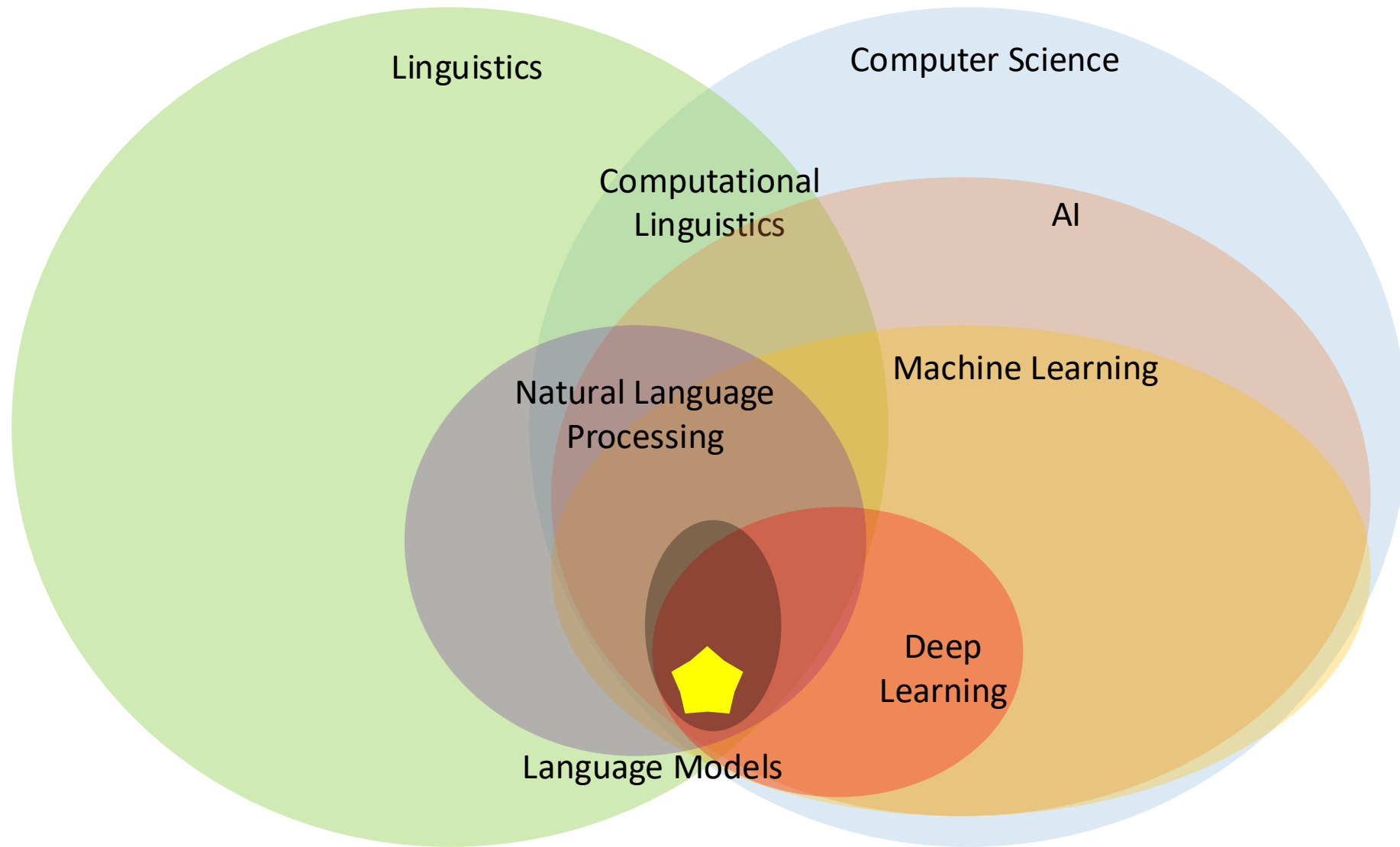
Natural Language Processing



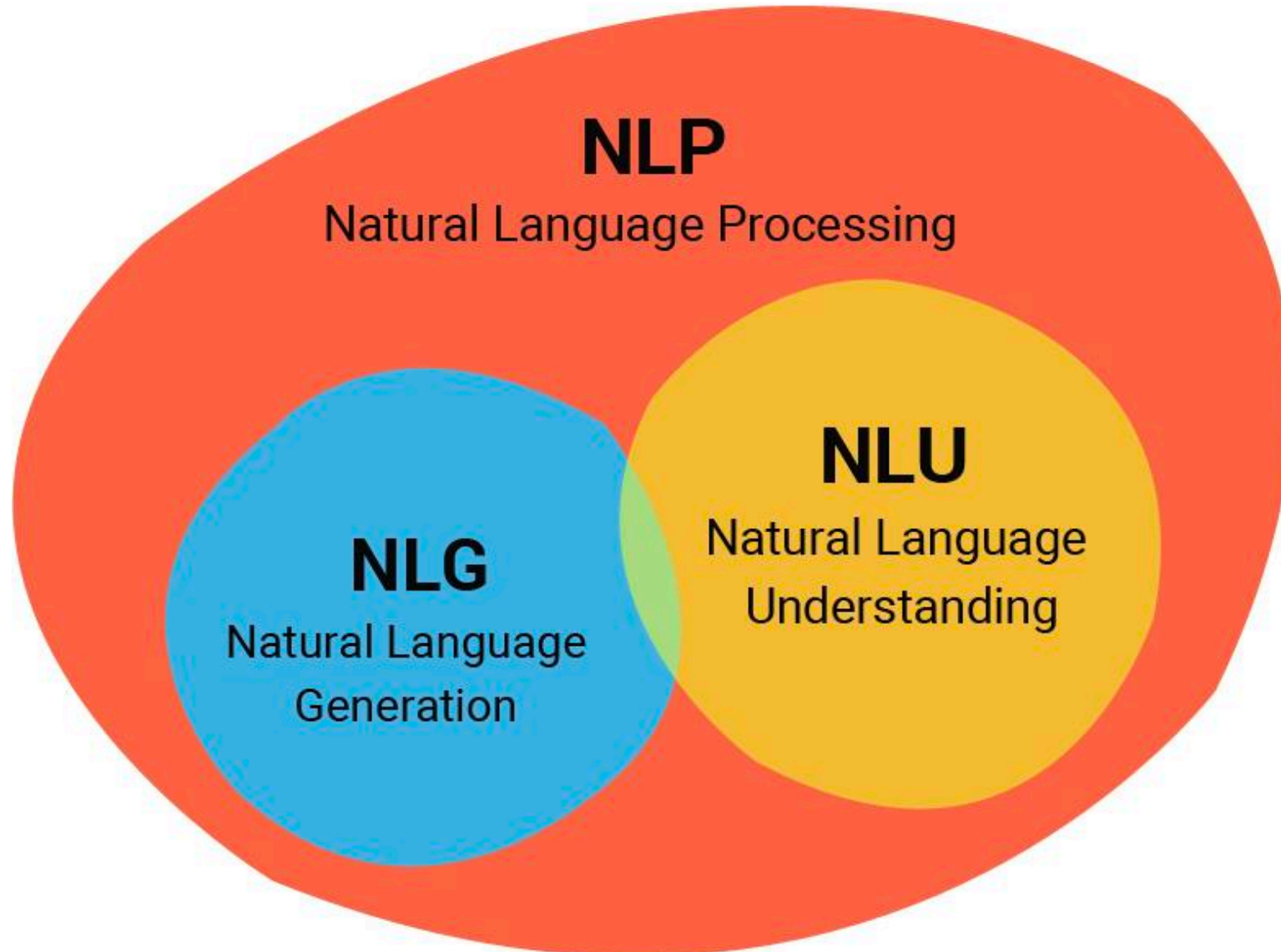
Natural Language Processing



Natural Language Processing



NLP/NLU/NLG



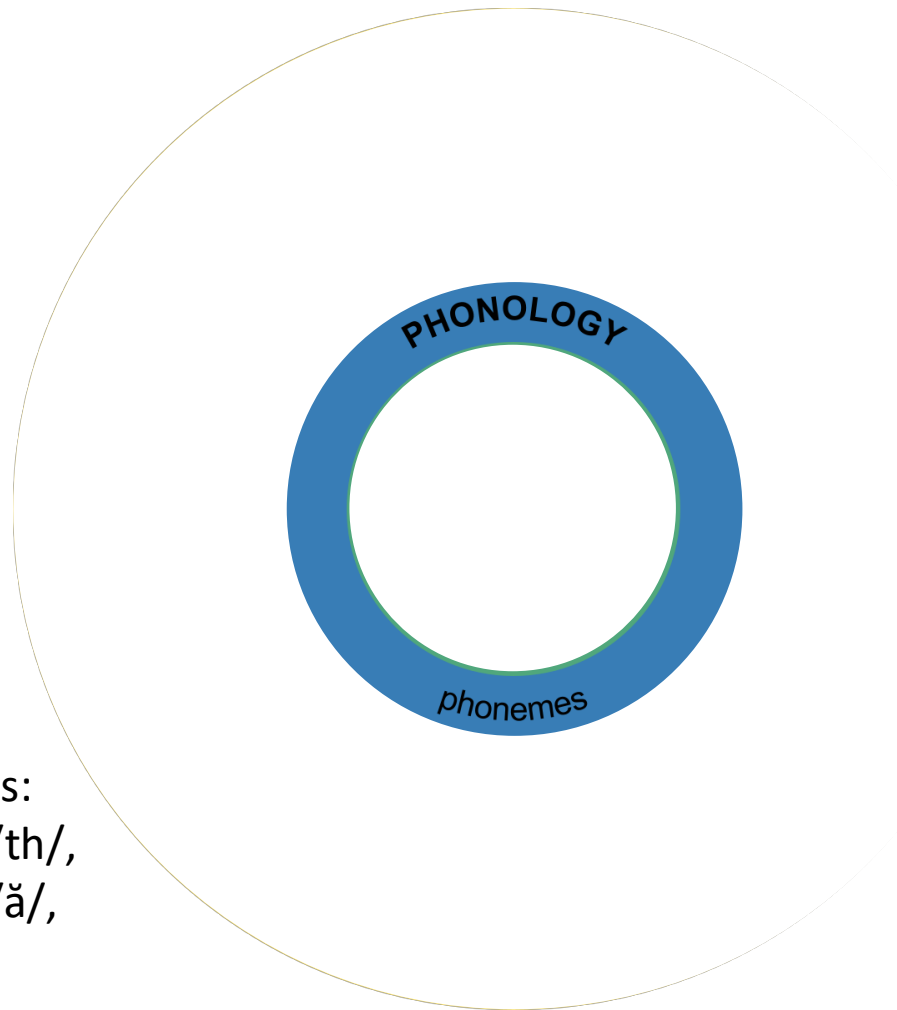
Components of Language

THE ENGLISH PHONOTYPIC ALPHABET.

VOWELS.				CONSONANTS.			
No.	Type.	Example of its sound.		No.	Type.	Example of its sound.	
LONG.							
1	E	e	eel	20	P	p	rop
2	A	a	ale	21	B	b	rob
3	Ä	a	alms	22	T	t	fate
4	Ö	o	all	23	D	d	fade
5	O	o	ope	24	C	c	etch
6	U	u	food	25	J	j	edge
SHORT.							
7	I	i	ill	26	C	c	leek
8	E	e	ell	27	G	g	league
9	A	a	am	28	F	f	safe
10	O	o	olive	29	V	v	sare
11	U	u	up	30	T	t	wreath
12	U	u	foot	31	H	h	wreath
DIPHTHONGS.							
13	Ä	i	isle	32	S	s	his
14	Ö	o	oil	33	Z	z	his
15	Ö	o	owl	34	S	f	vicious
16	U	u	male	35	K	g	vision
COALESCENTS.							
17	Y	y	yea	36	R	r	for
18	W	w	way	37	L	l	fall
ASPIRATE.							
19	H	h	hay	38	M	m	seem
				39	N	n	seen
				40	U	u	sing

wiki

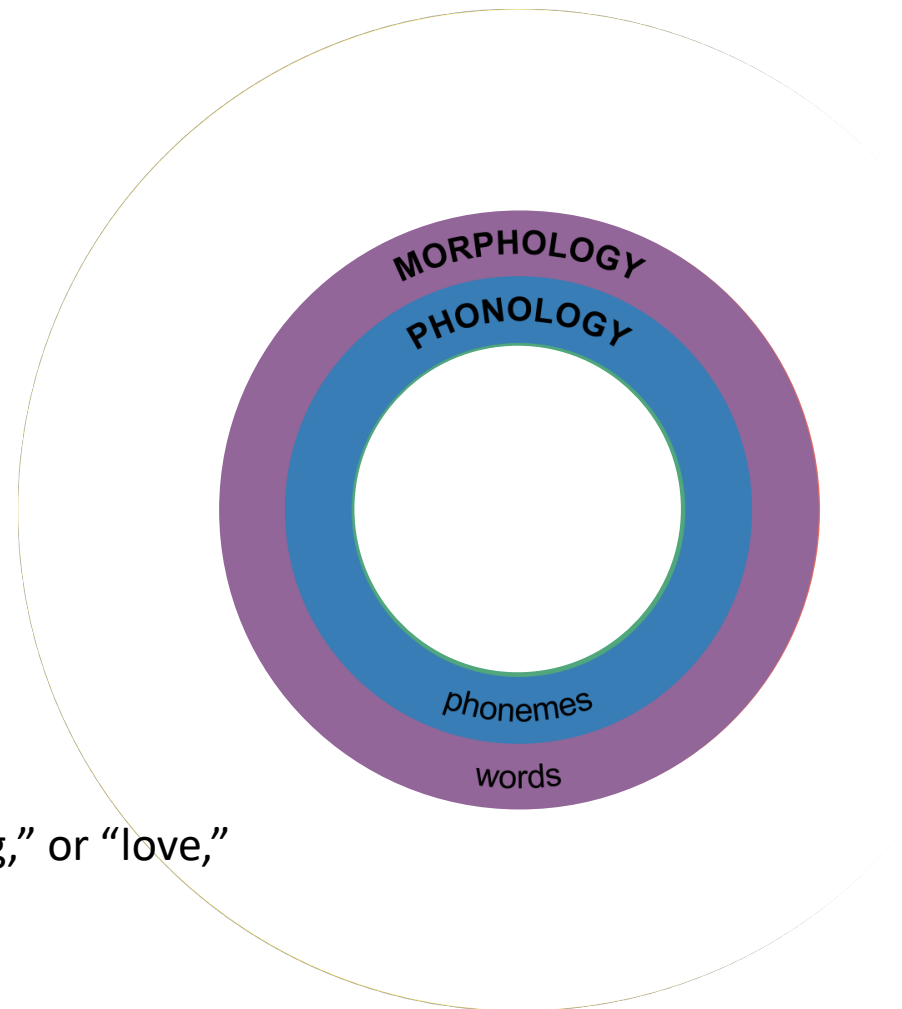
“that” contains three phonemes:
 “th” represents one phoneme /th/,
 “a” maps to the short a sound /ă/,
 “t” to its basic sound /t/.



study of speech structure
within a language

wiki

Components of Language



study of the smallest units of meaning, *morphemes*

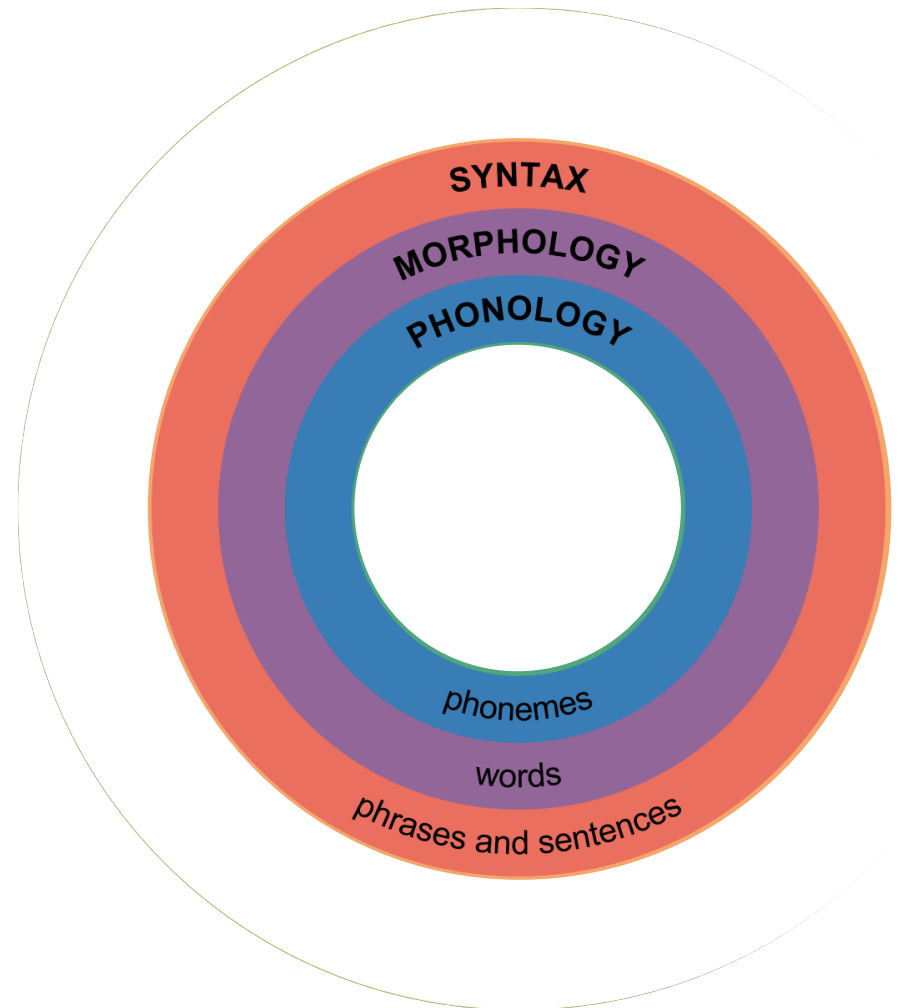
study of speech structure within a language

Morphemes:

base words, such as “hat,” “dog,” or “love,”
affixes, such as “un-,” “re-,”
plural “s” or “es,”
past tense “ed.”

Components of Language

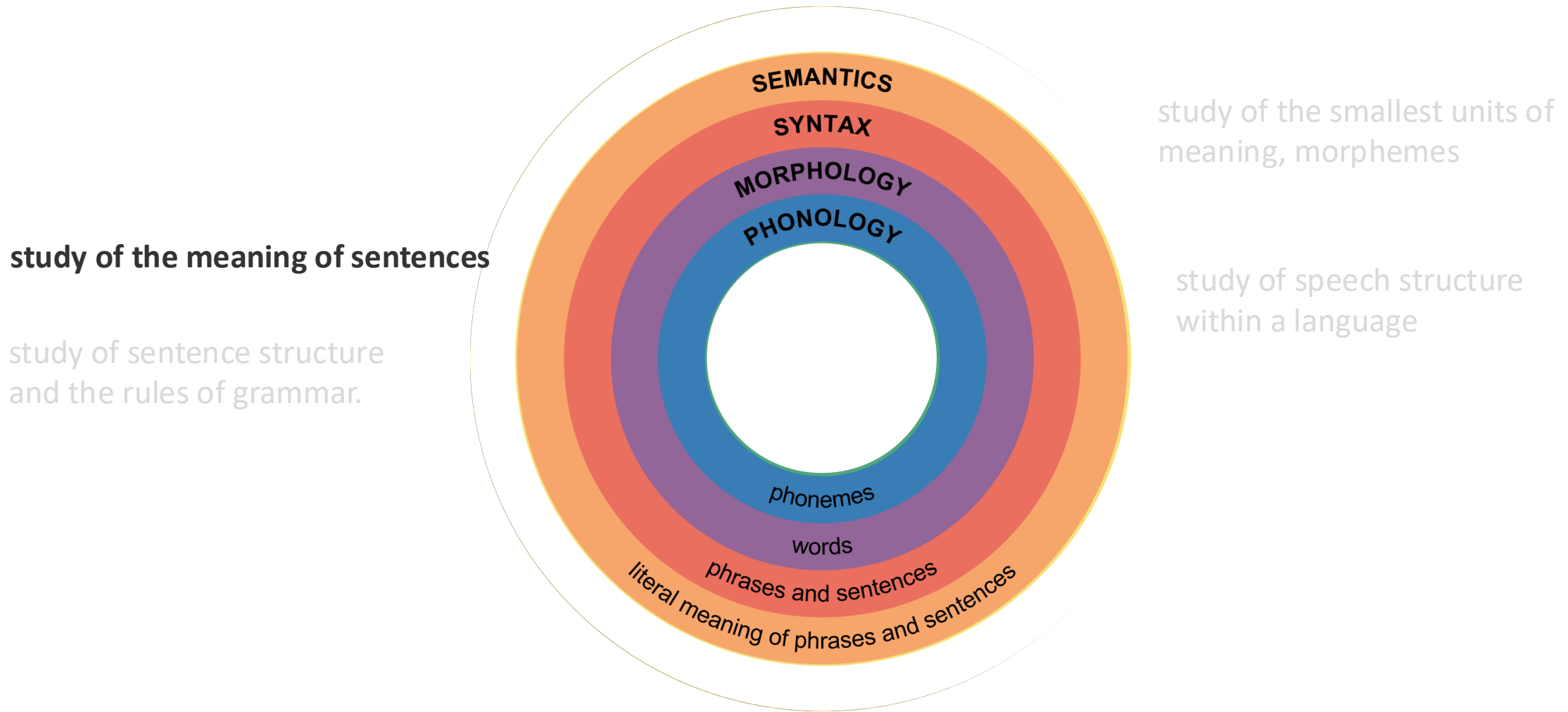
**study of sentence structure
and the rules of grammar.**



study of the smallest units of
meaning, morphemes

study of speech structure
within a language

Components of Language

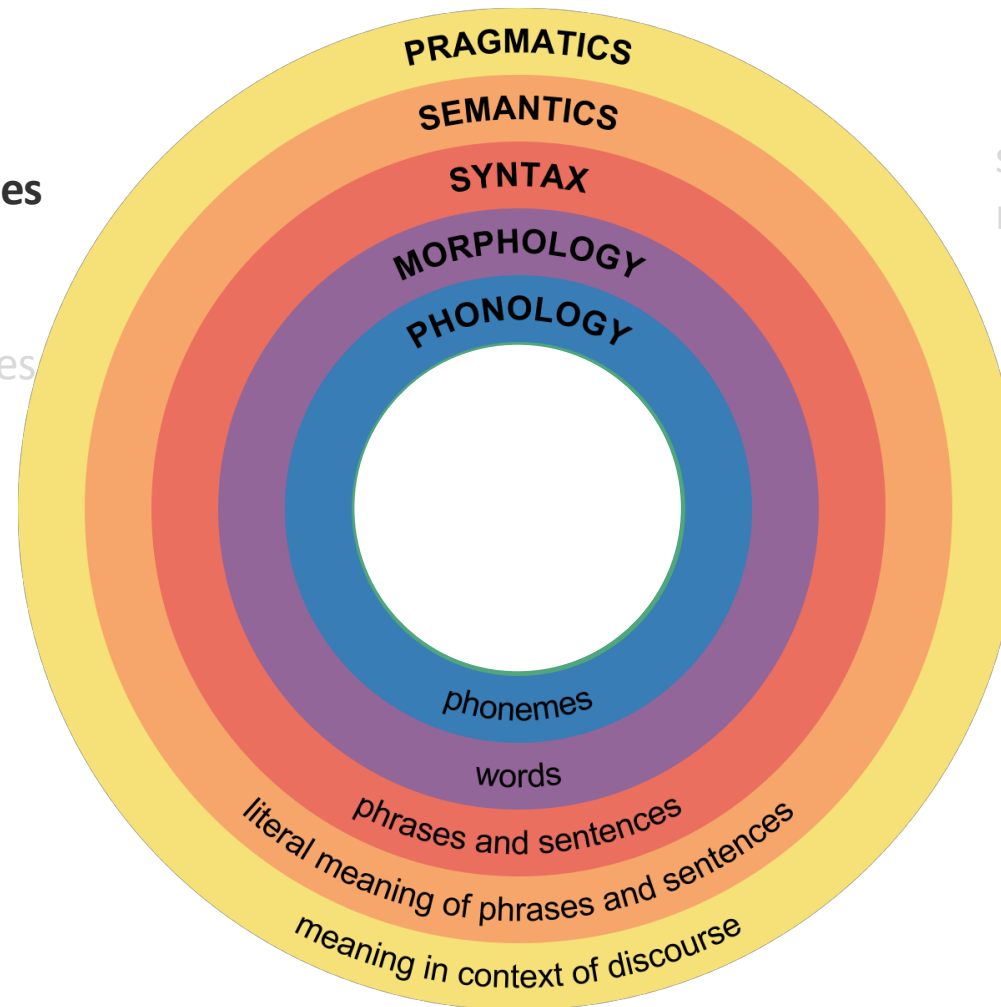


Components of Language

**study of the meaning of sentences
within a certain context**

study of the meaning of sentences

study of sentence structure
and the rules of grammar.



study of the smallest units of
meaning, morphemes

study of speech structure
within a language

The interaction of
semantic knowledge
with our knowledge
of the world

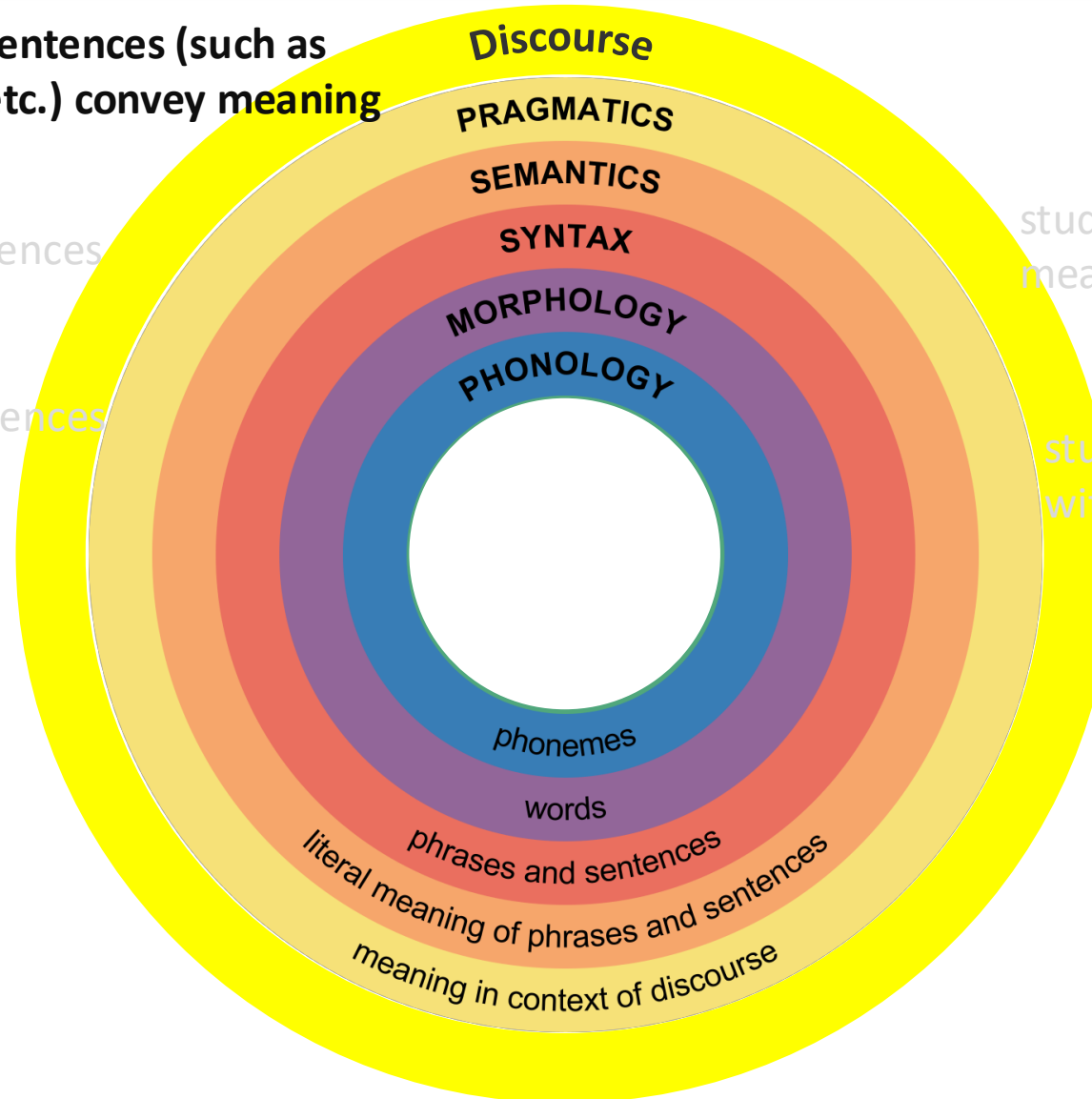
Components of Language

study of how sequences of sentences (such as conversations, paragraphs, etc.) convey meaning and information

study of the meaning of sentences within a certain context

study of the meaning of sentences

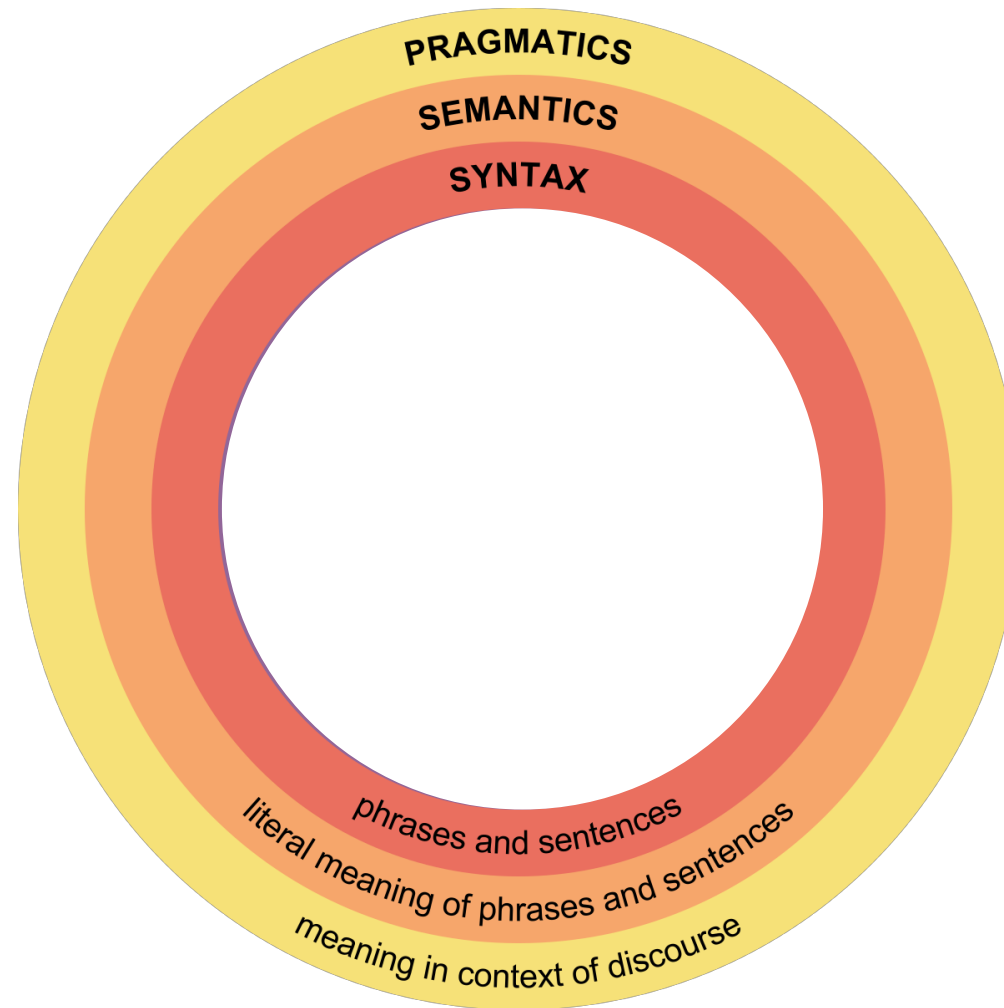
study of sentence structure and the rules of grammar.



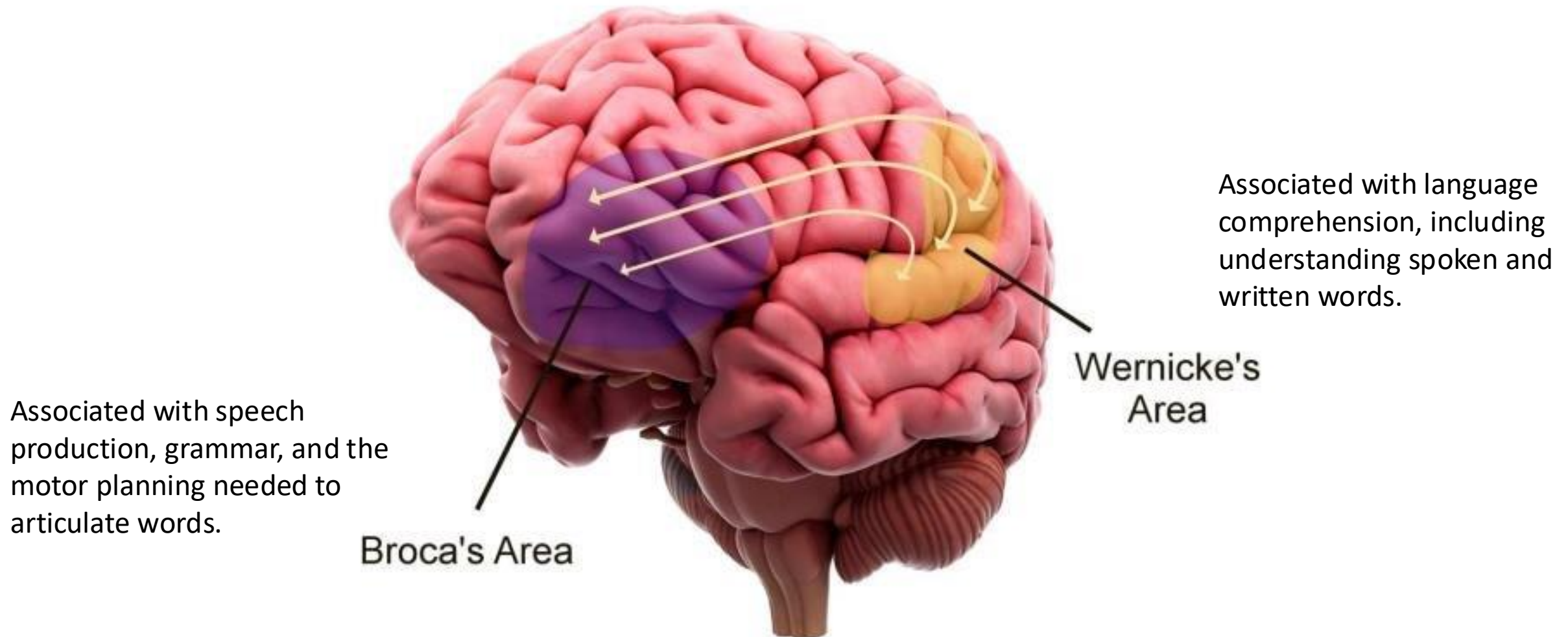
study of the smallest units of meaning, morphemes

study of speech structure within a language

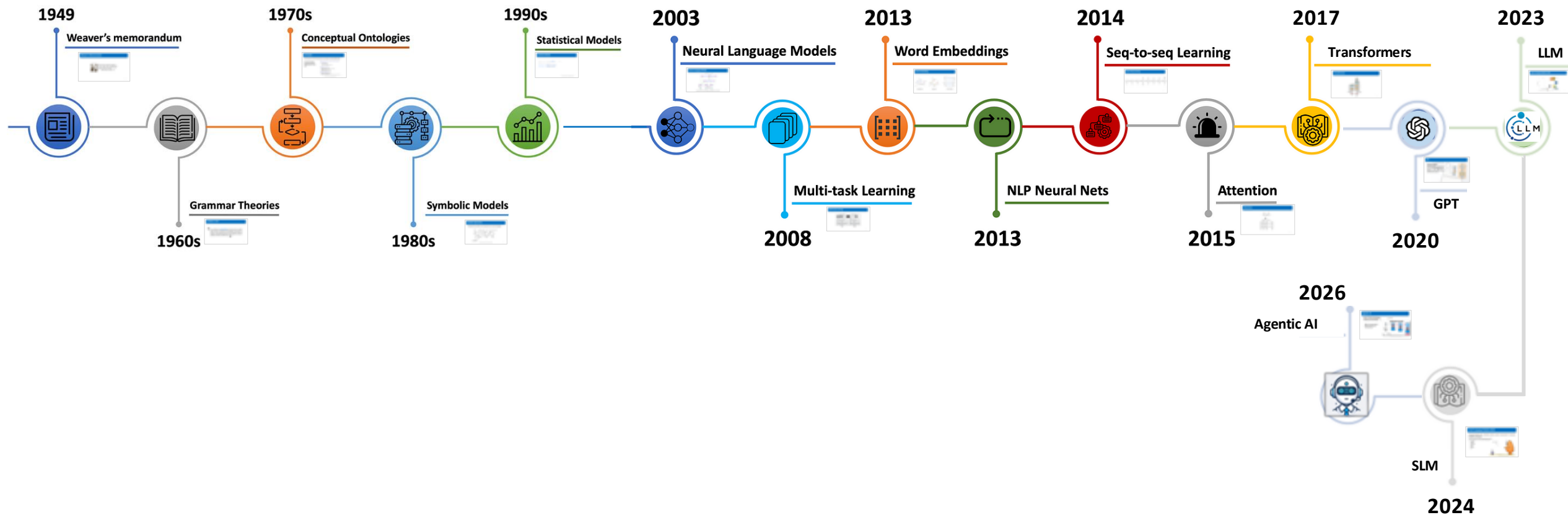
Components of Language - NLP



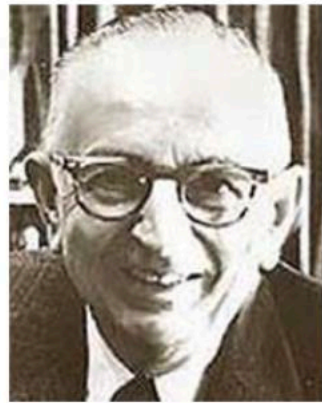
Anatomy of Natural Language



History of NLP



Weaver's Memorandum



“When I look at an article in Russian, I say: this is really written in English, but it has been coded in some strange symbols. I will now proceed to decode.”

- Warren Weaver, March 1947

Dictionary Lookup

Dictionary: The ↔ Il
 Red ↔ Rosso
 House ↔ Casa

Word order rules: Adjective + Noun ↔ Noun + Adjective

The red house

Dictionary lookup

Il rosso casa

Il casa rosso

Reorder words

La casa rossa

But it should be...

AI Winter in NLP

“ As a conclusion, the [recommendations](#) of the report were to support machine translation research less on the achievement of a full end-to-end solution and more on being able to aid translators in their work. For example, the report suggested to look for practical methods for evaluating translations, evaluation of available sources of translations, and ways of speeding up the human translation process. ”

Grammar

Welcome to

EEEEEE	LL	IIII	ZZZZZZ	AAAAA
EE	LL	II	ZZ	AA AA
EEEEEE	LL	II	ZZZ	AAAAAAA
EE	LL	II	ZZ	AA AA
EEEEEE	LLLLLL	IIII	ZZZZZZ	AA AA

Eliza is a mock Rogerian psychotherapist.

The original program was described by Joseph Weizenbaum in 1966.

This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?

YOU: Men are all alike.

ELIZA: What is the connection, do you suppose ?

YOU: They're always bugging us about something or other.

ELIZA: Can you think of a specific example ?

YOU: Well, my boyfriend made me come here.

ELIZA: Is it important to you that your boyfriend made you come here ?

YOU: He says I'm depressed much of the time.

ELIZA: I am sorry to hear that you are depressed.

YOU: It's true. I am unhappy.

ELIZA: Can you explain what made you unhappy ?

YOU:

Ontologies

Structuring real-world information into computer-understandable data

Conceptual Ontologies:

Hand-built hierarchies of concepts (objects, actions, goals)

MARGIE: Memory, Analysis, Response Generation in English

the system combined a

- parser (English $\rightarrow$ CD)
- generator (CD $\rightarrow$ English)
- inference engine (inferred info from CD)

MARGIE in inference mode

INPUT: John gave Mary an aspirin.

OUTPUT1: John believes that Mary wants an aspirin.

OUTPUT2: Mary is sick.

OUTPUT3: Mary wants to feel better.

OUTPUT4: Mary will ingest the aspirin.

MARGIE in paraphrase mode

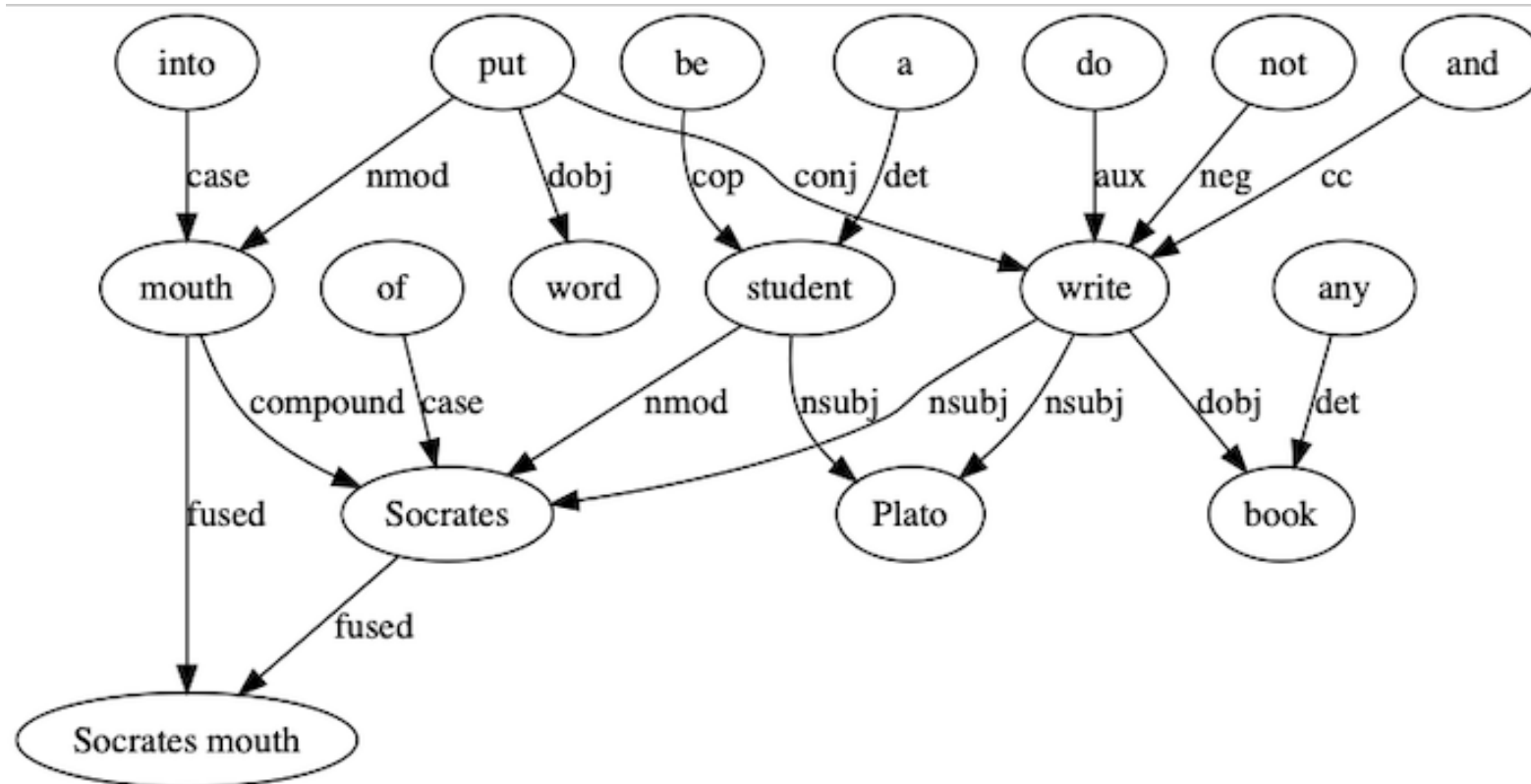
INPUT: John killed Mary by choking her.

OUTPUT1: John strangled Mary.

OUTPUT2: John choked Mary and she died because she could not breathe.

Symbolic Approaches

Complex hard-coded rules and grammars to parse language



Statistical Models

$$P(\mathbf{x}^{(t+1)} | \mathbf{x}^{(t)}, \dots, \mathbf{x}^{(1)}) = P(\mathbf{x}^{(t+1)} | \overbrace{\mathbf{x}^{(t)}, \dots, \mathbf{x}^{(t-n+2)}}^{n-1 \text{ words}})$$

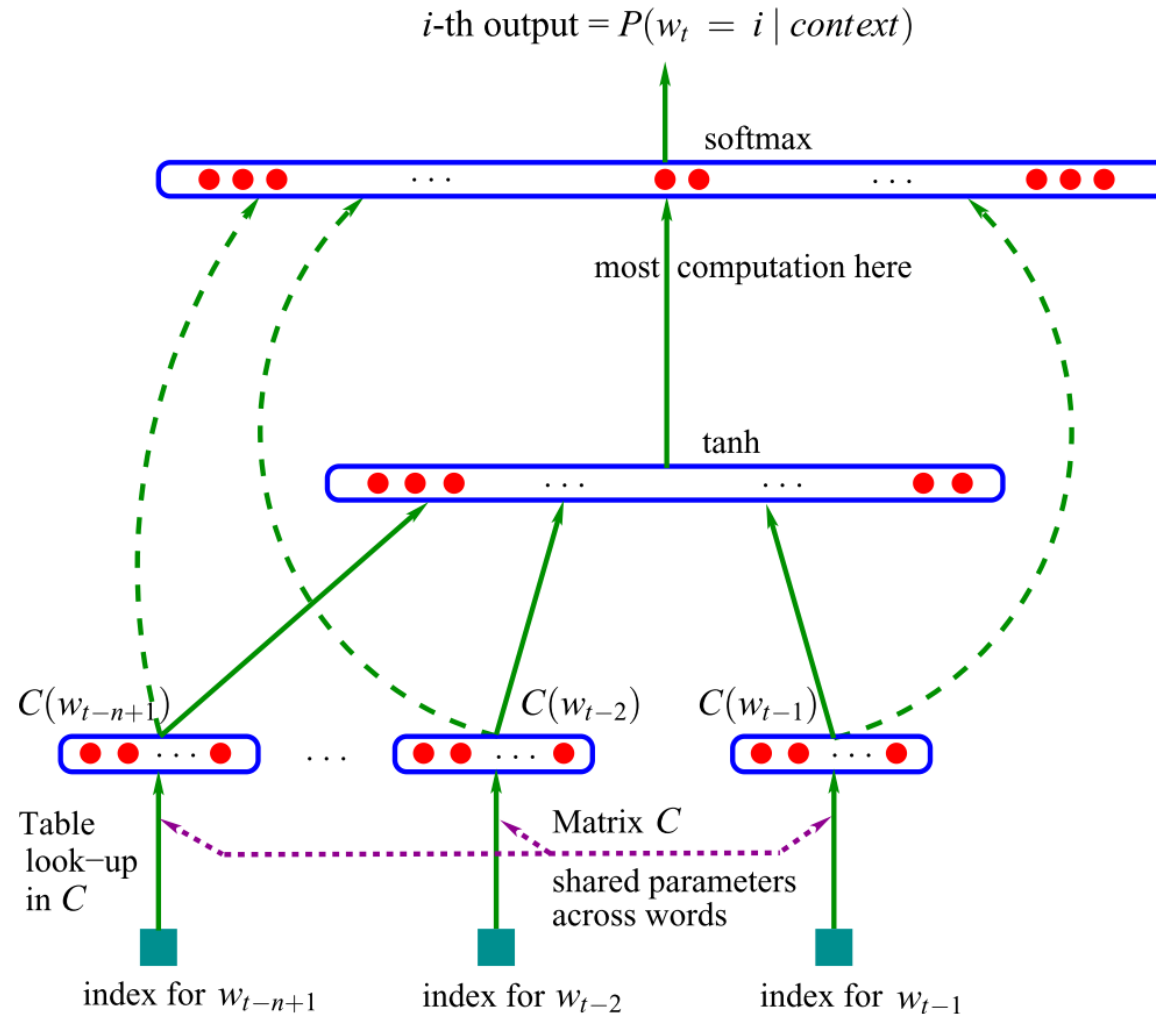
I _____

prob of a n-gram $\rightarrow$ $P(\mathbf{x}^{(t+1)}, \mathbf{x}^{(t)}, \dots, \mathbf{x}^{(t-n+2)})$

$=$

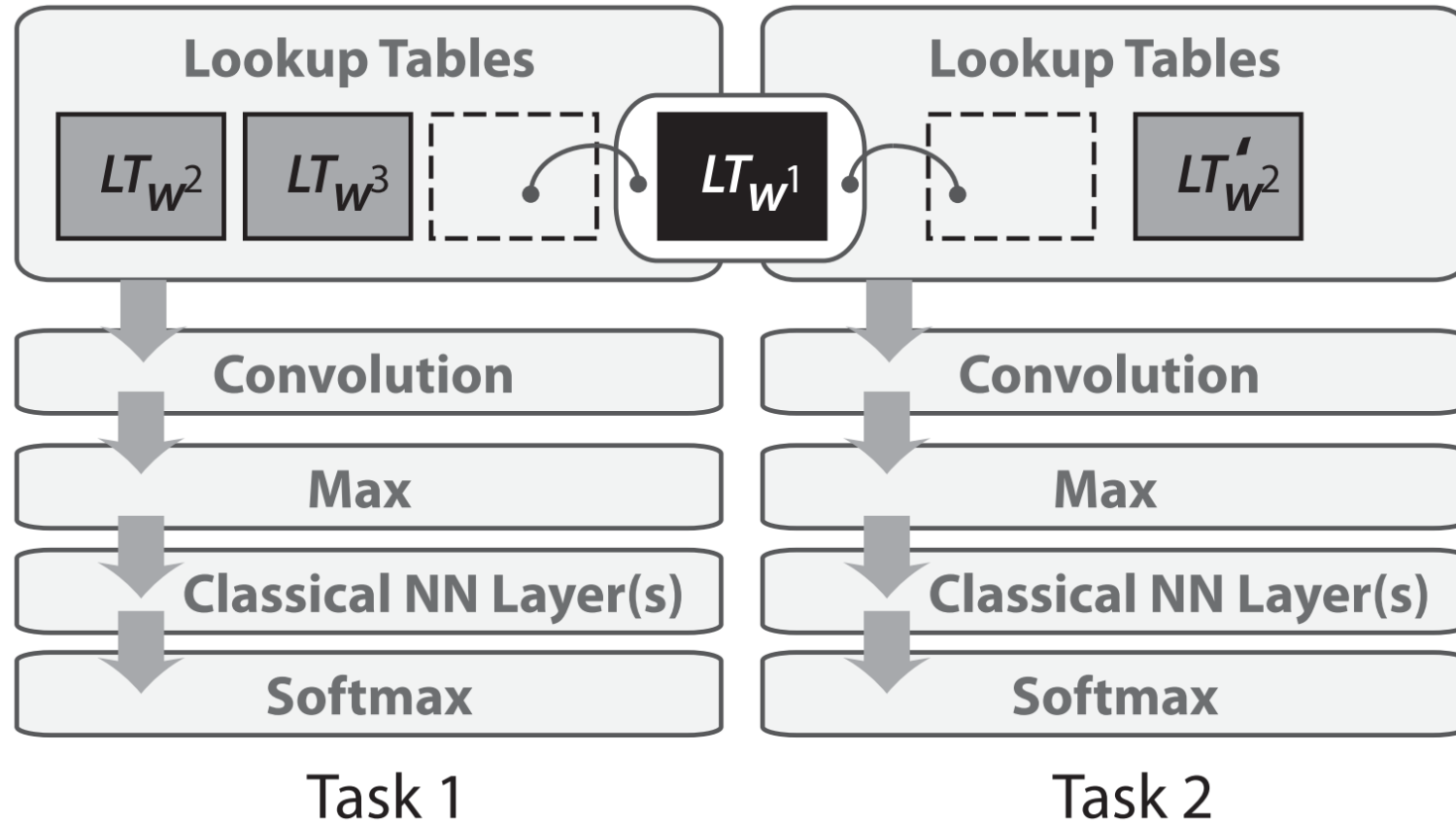
prob of a (n-1)-gram $\rightarrow$ $P(\mathbf{x}^{(t)}, \dots, \mathbf{x}^{(t-n+2)})$

Neural Language Models

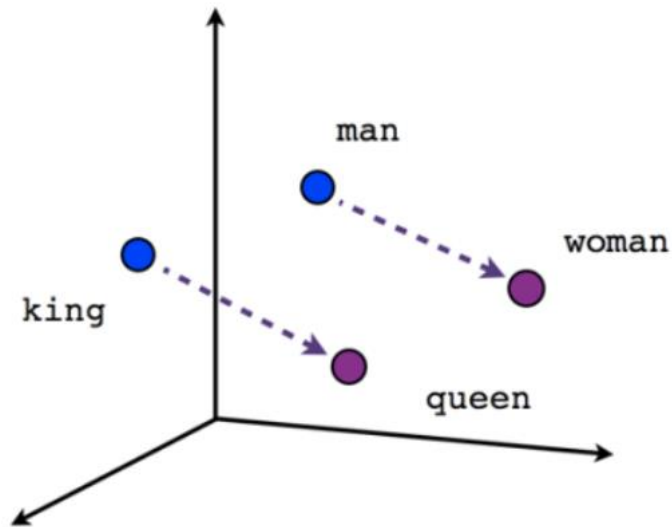


A feed-forward neural network language model (Bengio et al., 2001; 2003)

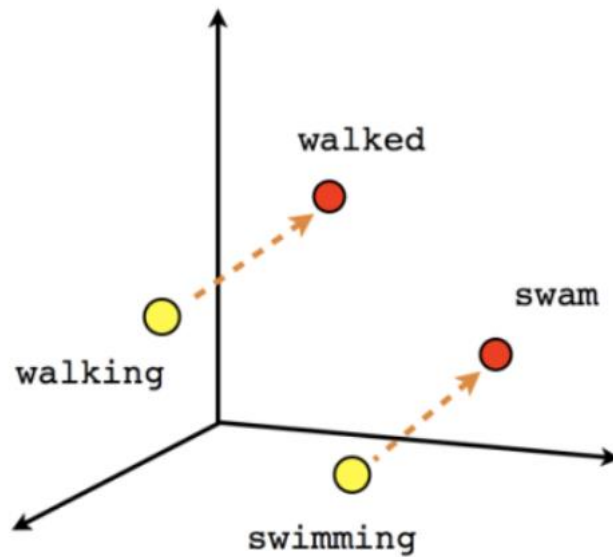
Multi-task Learning



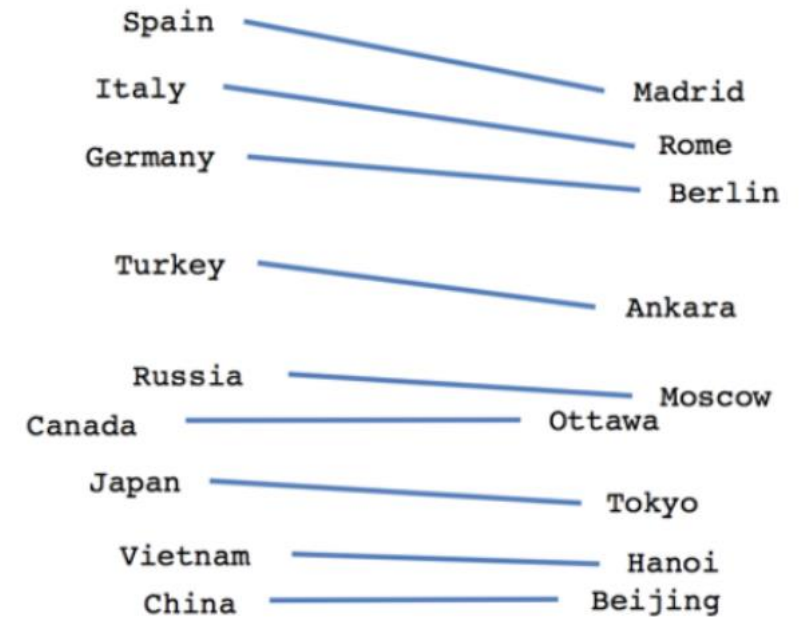
Word Embedding



Male-Female

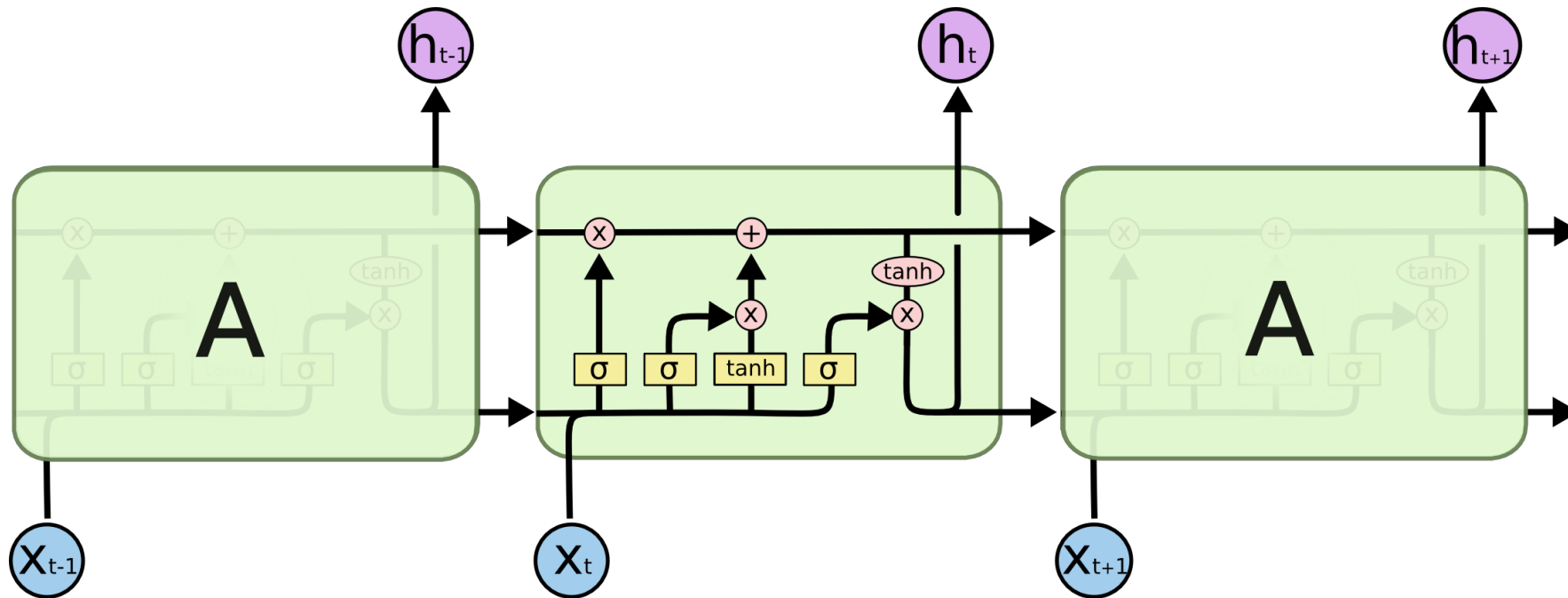


Verb tense

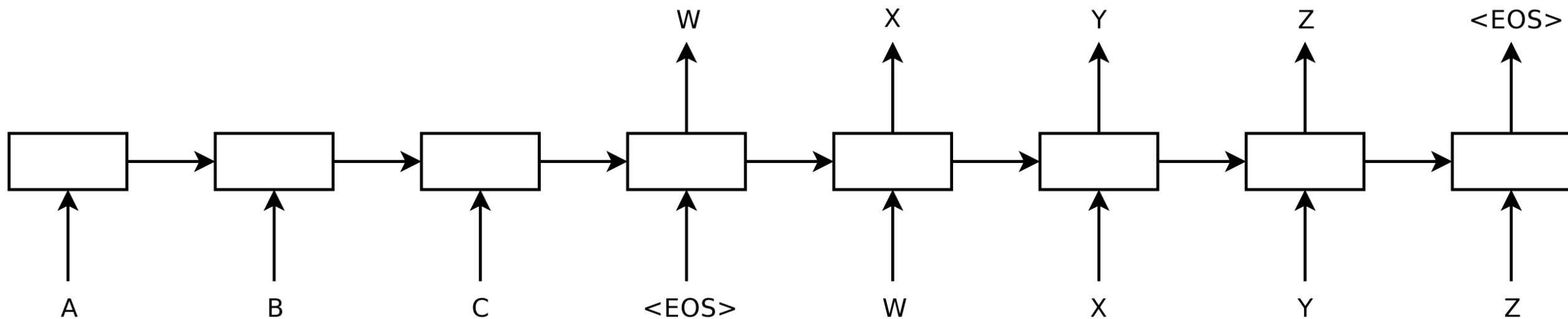


Country-Capital

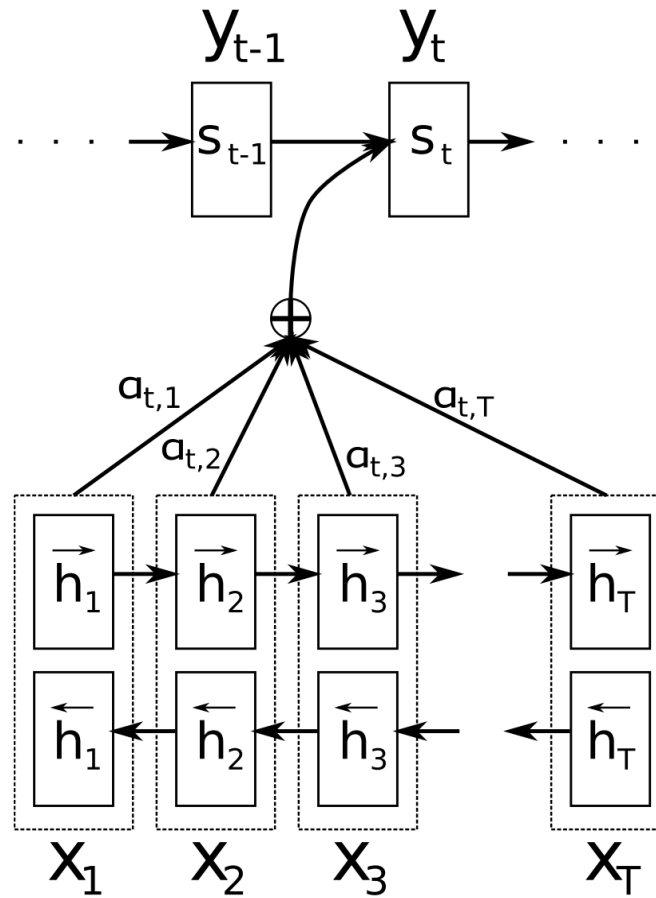
Recurrent Neural Networks (RNN)



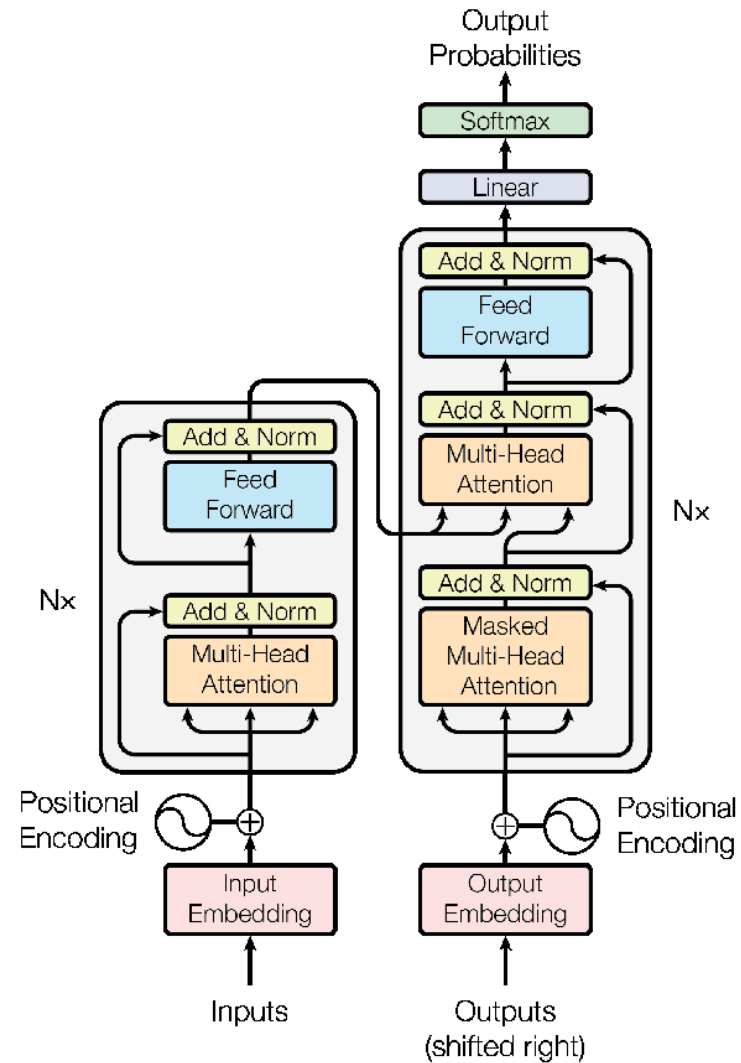
Seq-2-Seq Learning



Attention



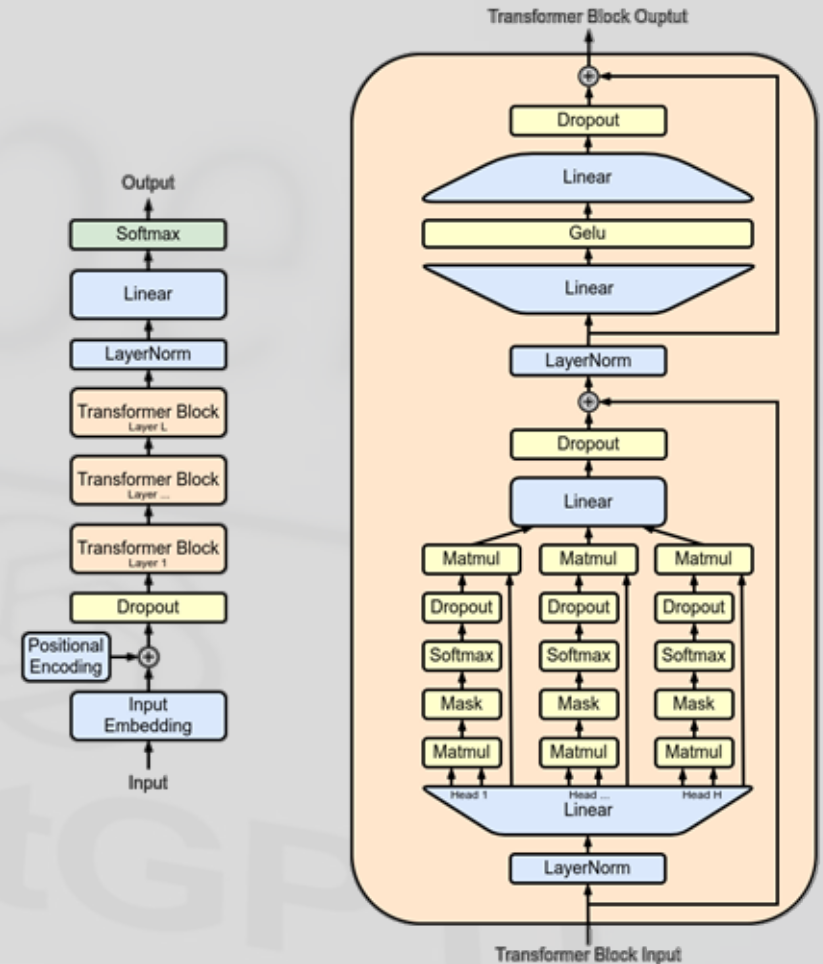
Transformers



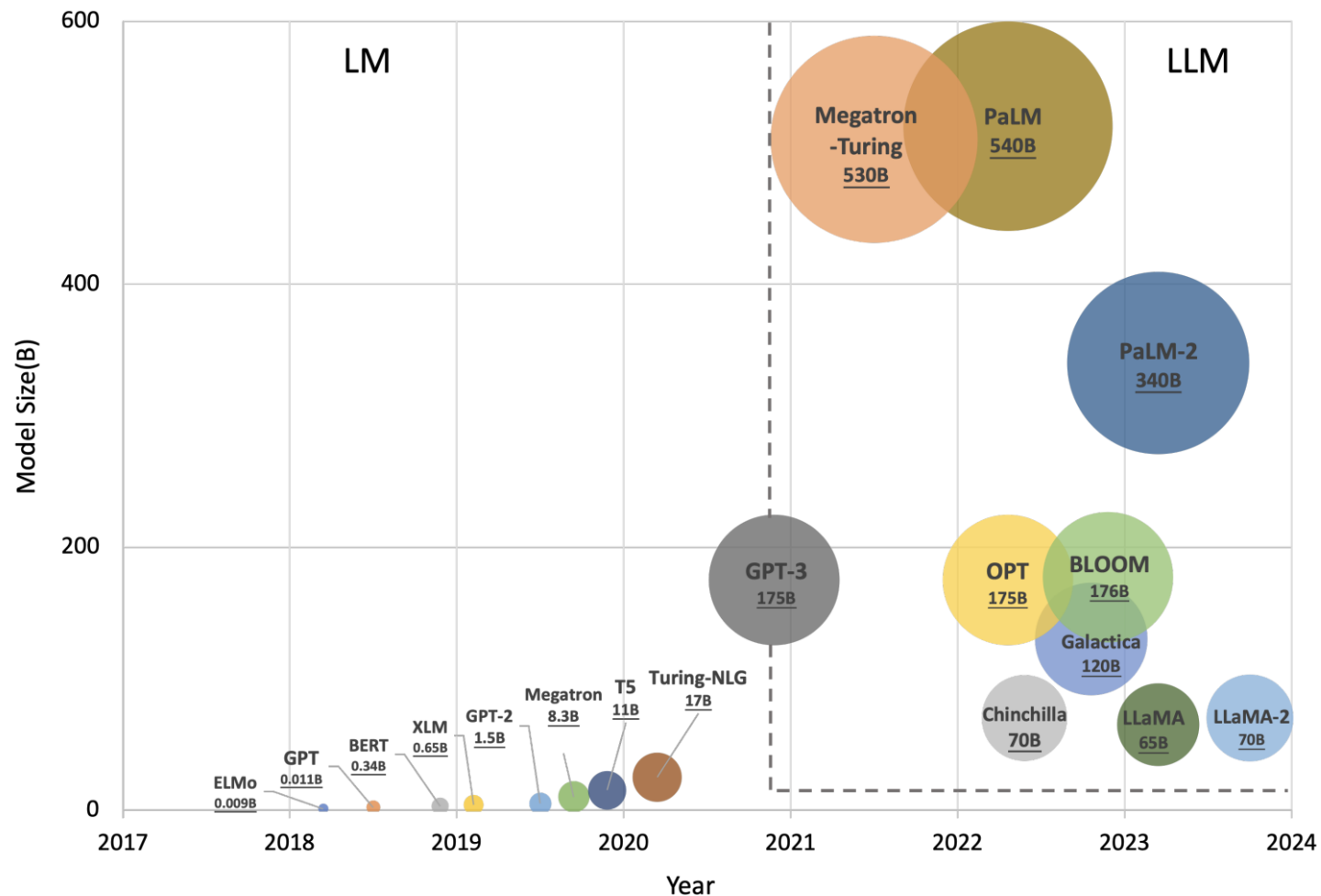
GPT

Generative pre-trained transformers (GPT)

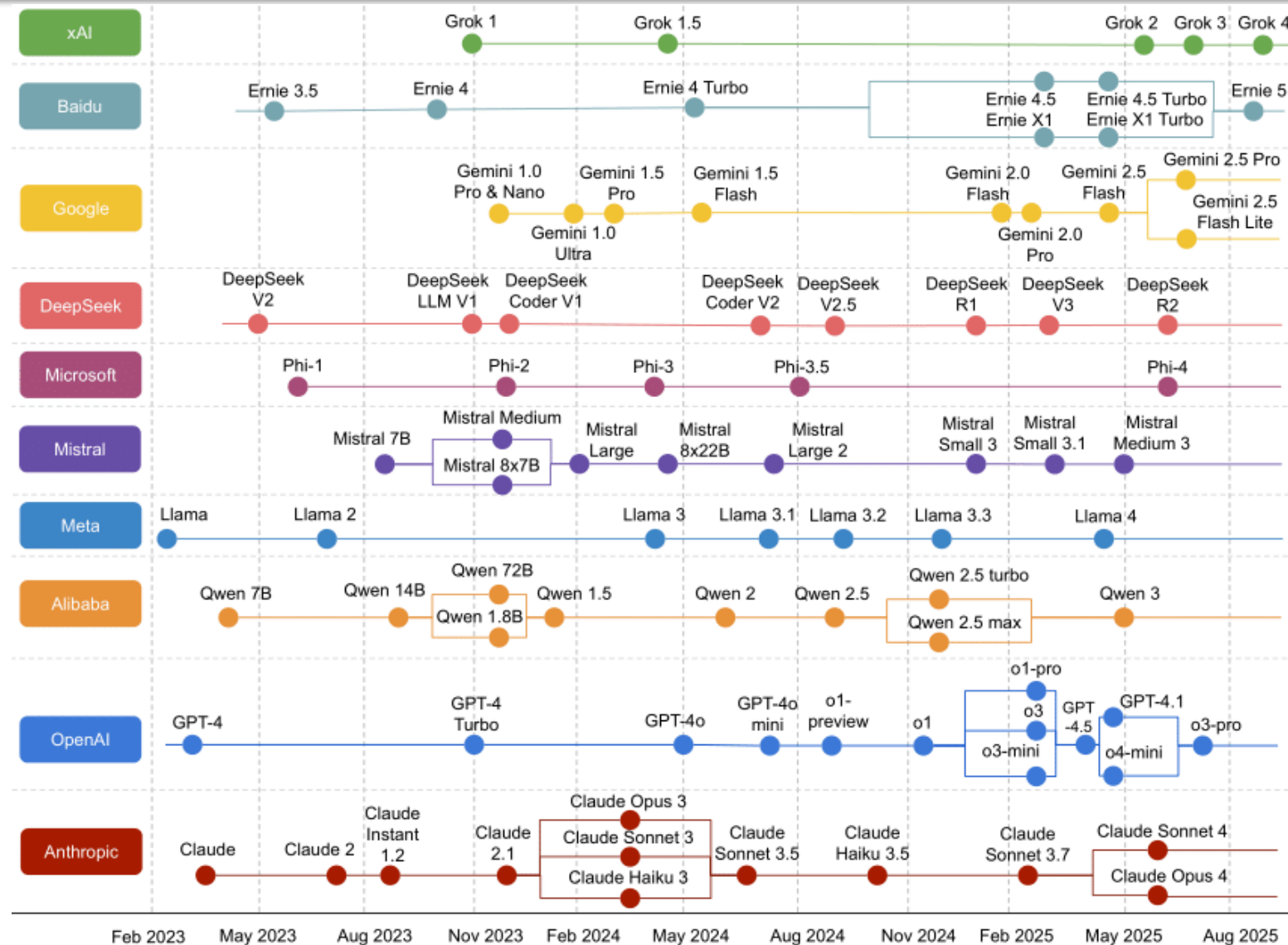
A type of large language model based on the transformer architecture, pre-trained on large data sets of unlabeled text, and able to generate novel human-like content.



Large Language Models (LLM)



Large Language Models (LLM)

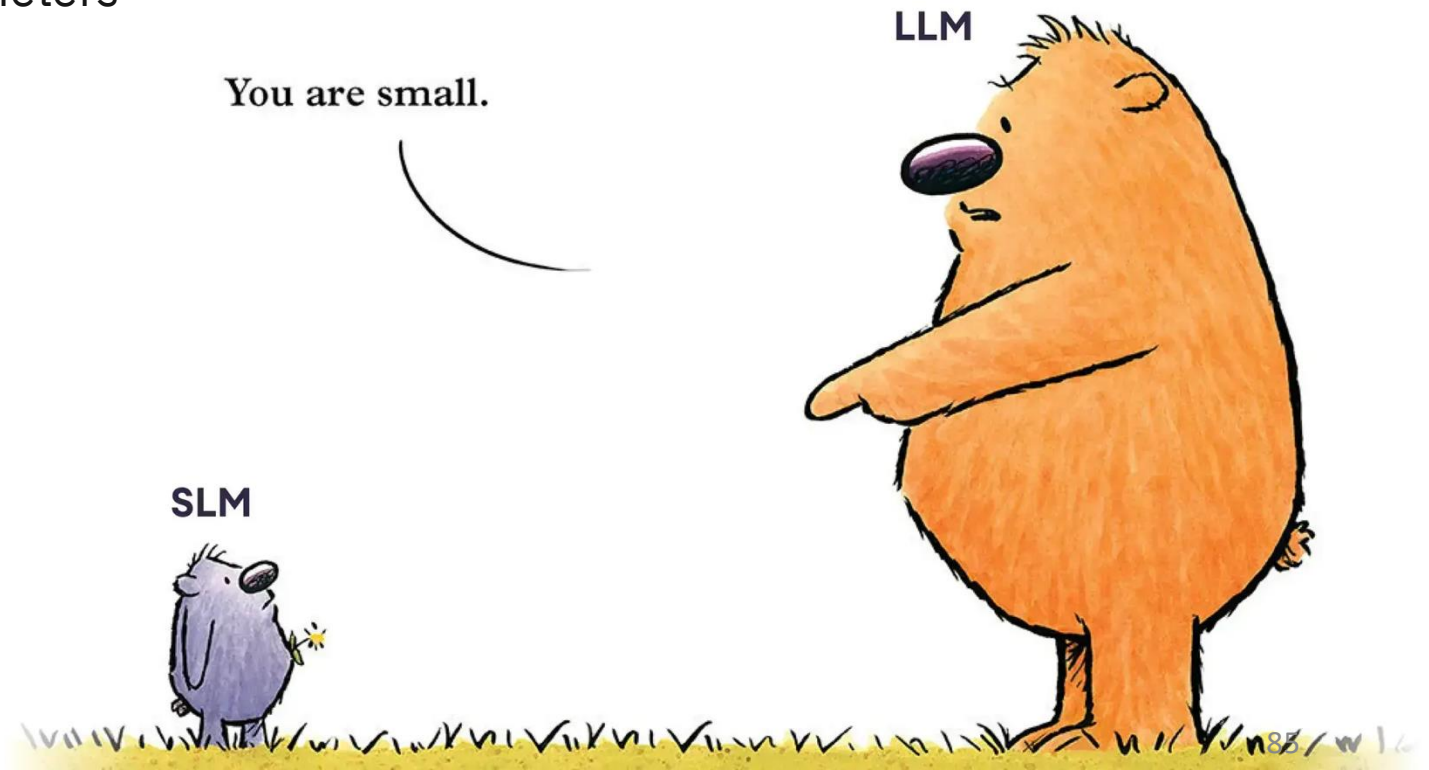


Small Language Models (SLM)

Language Models with a relatively modest number of parameters* compared to larger counterparts.

* a few million to a few billion, as opposed to LLMs with hundreds of billions or even trillions of parameters

- DistilBERT
- Gemma
- GPT-4o mini
- Granite
- Llama
- Ministral
- Phi

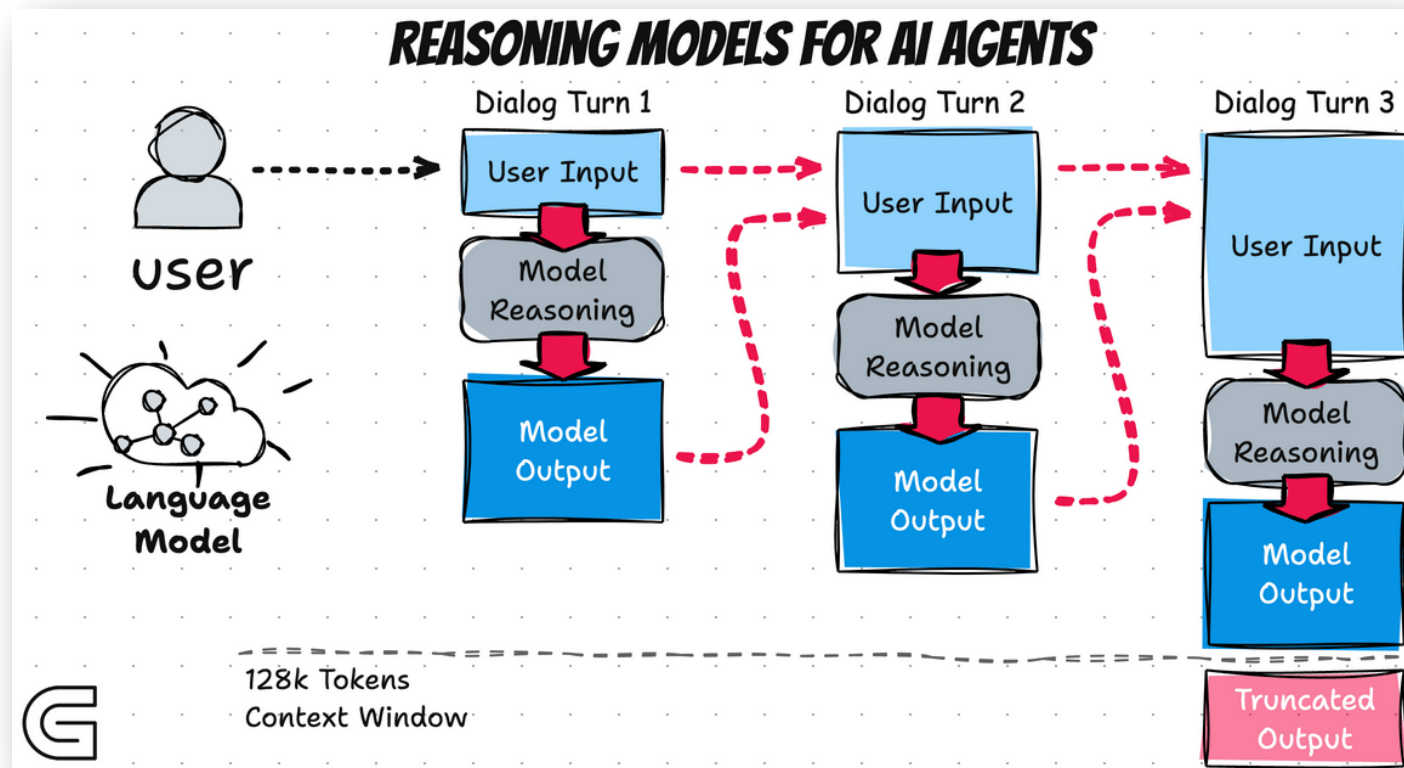


Agentic AI

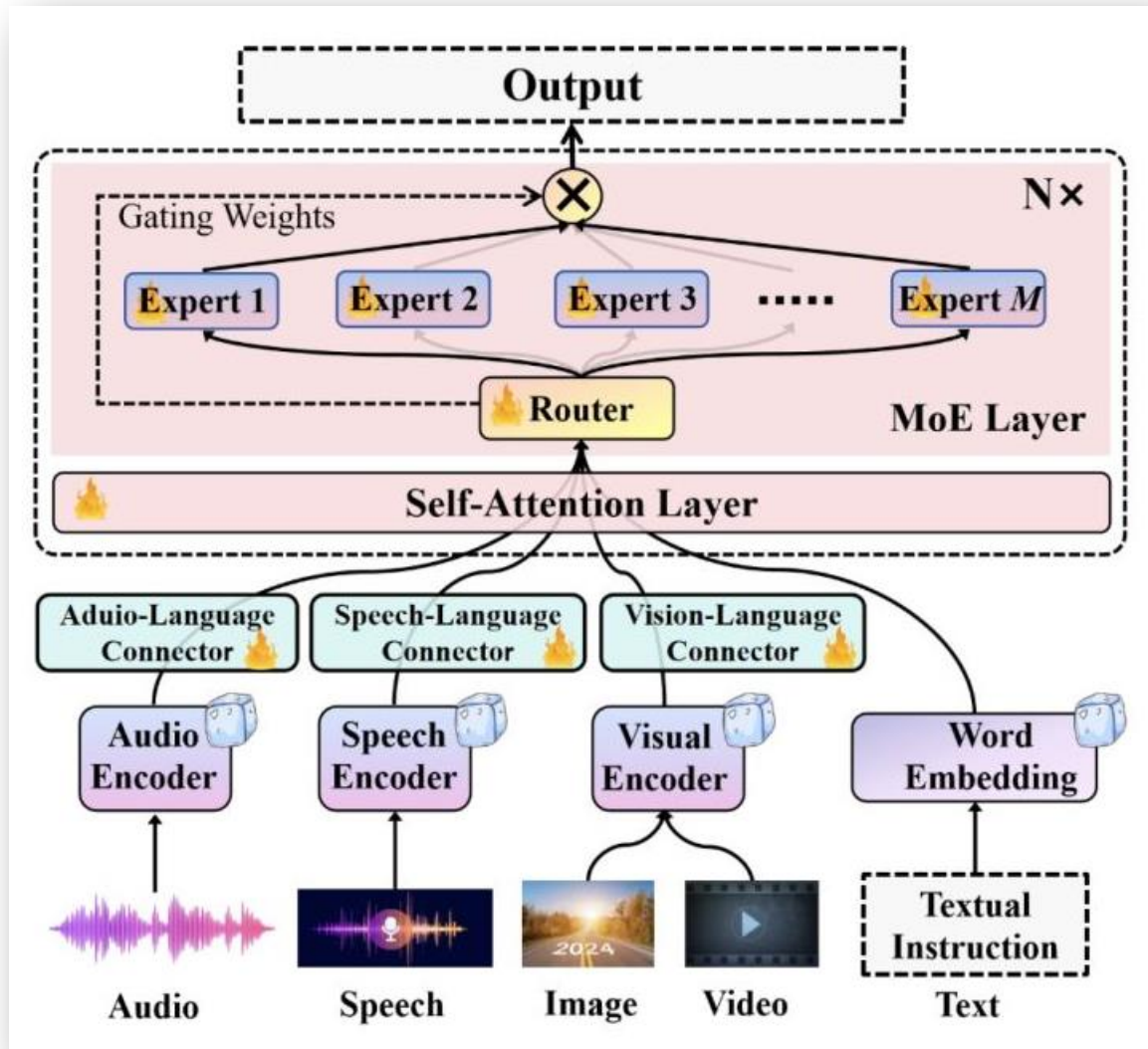
Shift to "Reasoning Models"
(Inference-Time Scaling)

Idea: Encourage models to generate intermediate reasoning steps.

e.g. GPT O1



Agentic AI

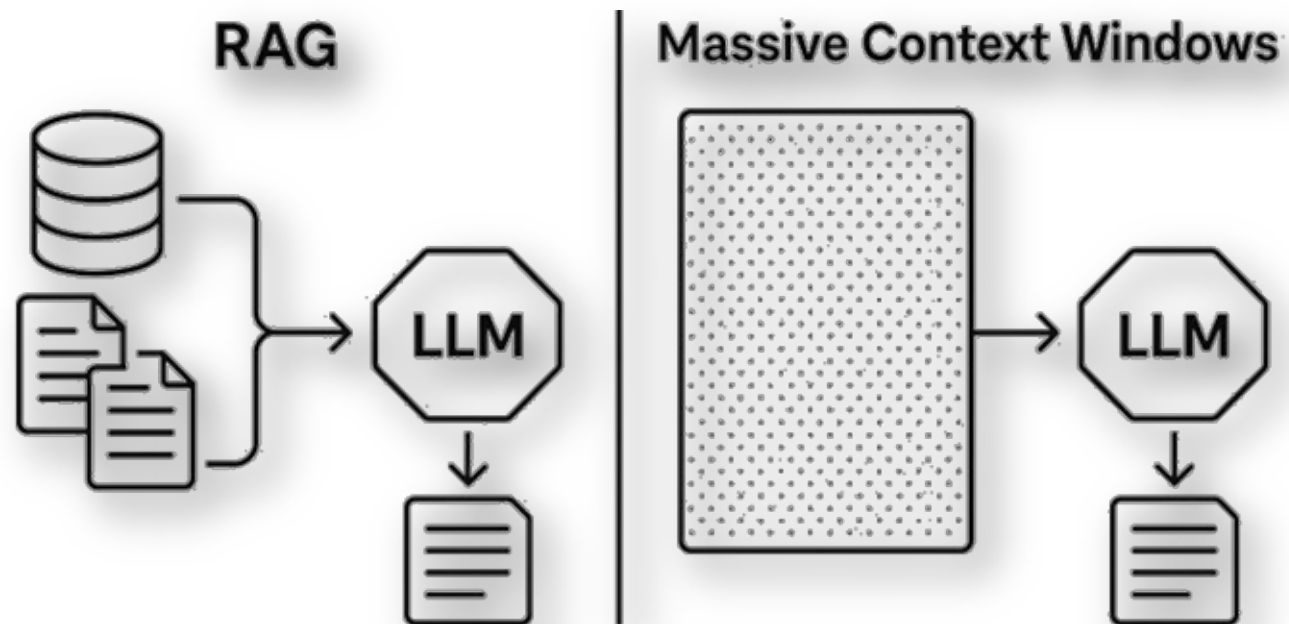
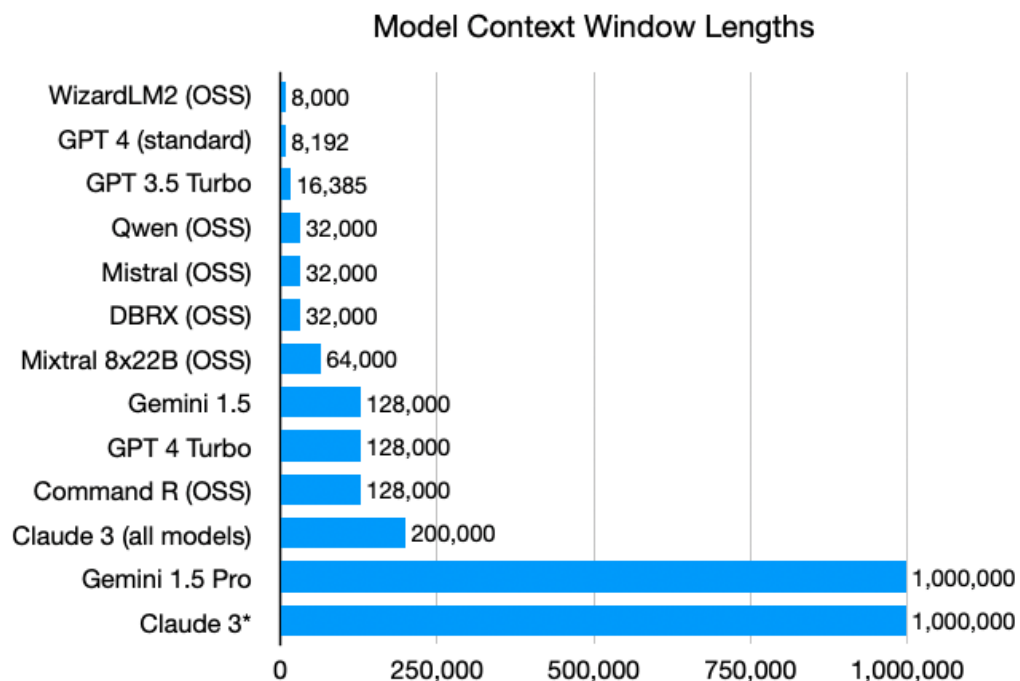


Architectural Innovations:
Sparse & Multimodal

"Omni" models
e.g. GPT 4o

Agentic AI

The "Context Revolution" (Long Context vs. RAG)



<https://medium.com/@mustafa.gencc94/rag-vs-massive-context-windows-is-retrieval-augmented-generation-really-obsolete-856db7812751>

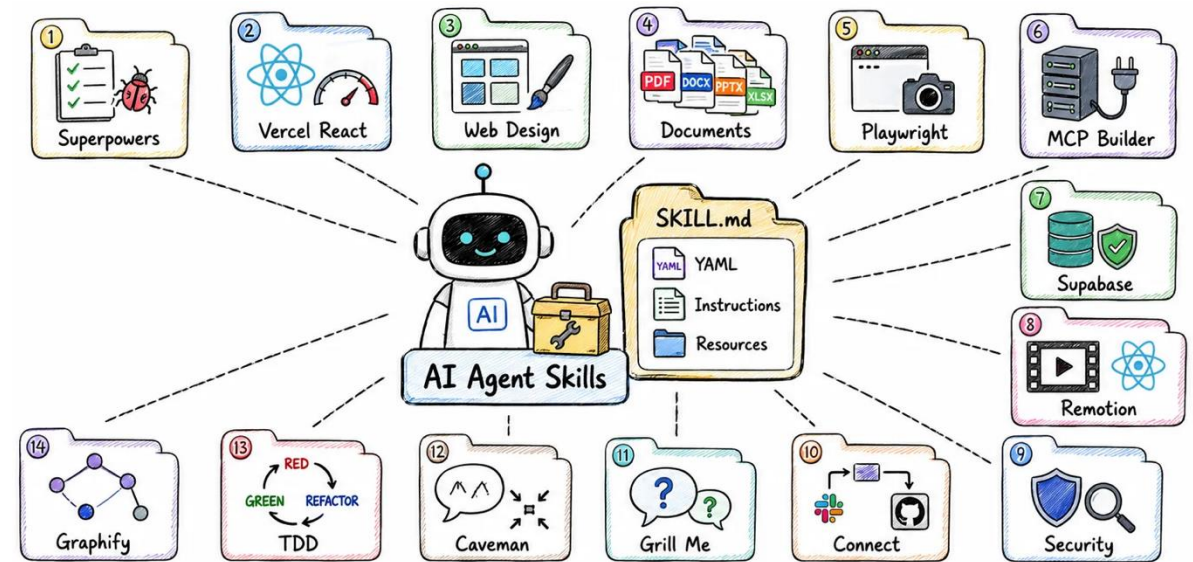
Agentic AI

From Models to Skills

An AI agent is more than an LLM:
it combines capabilities to accomplish a goal.

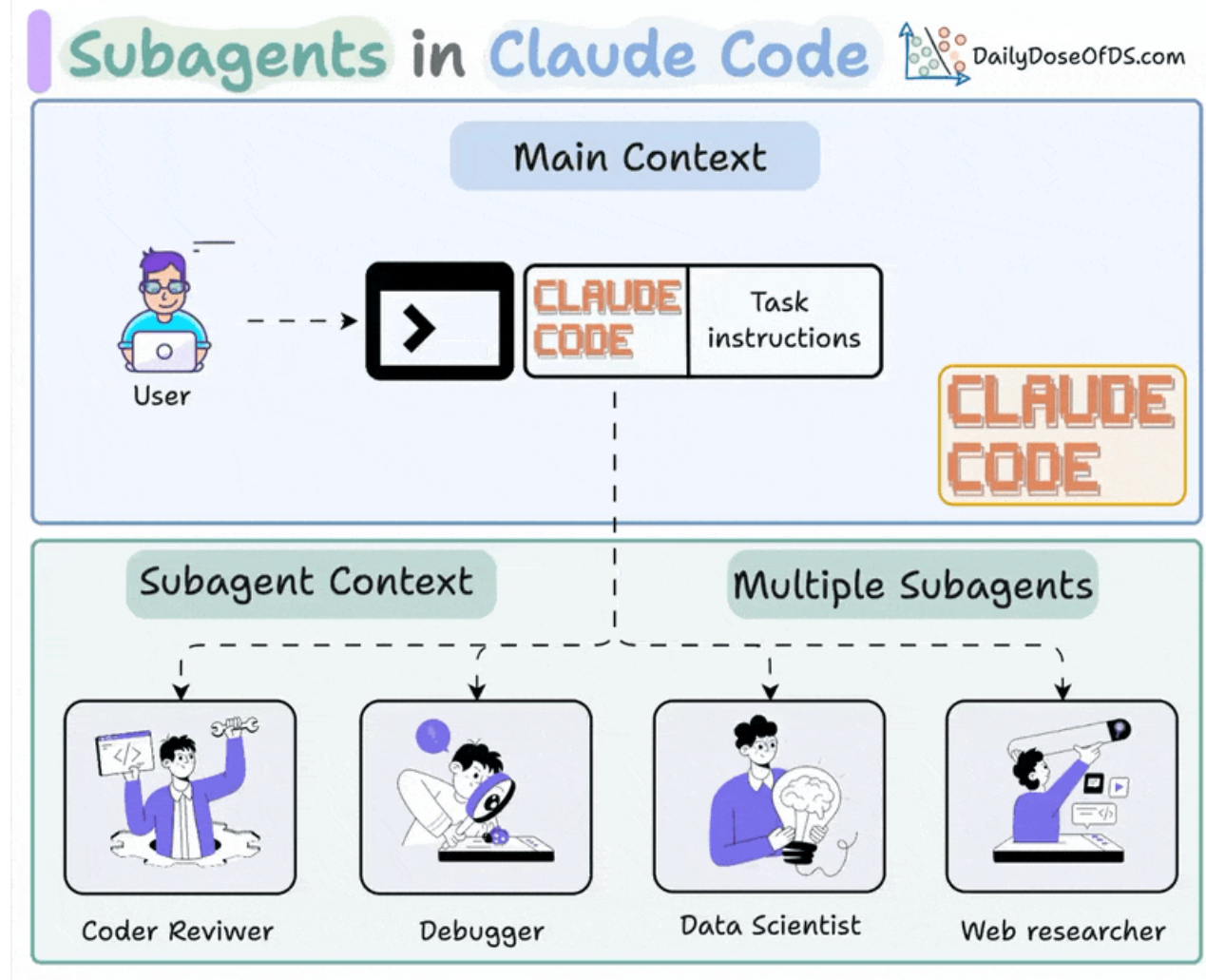
Skills are self-contained folders of instructions and resources that AI agents load on demand to gain specialized expertise

Skills turn an LLM from a “text generator” into a goal-directed system.



https://pinggy.io/blog/ai_agent_skills/

Agentic AI



From Chatbots to
"Agentic Workflows"

Challenges of Natural Language Processing

Productivity

Ambiguous

Variability

Diversity

Sparsity

Definition “property of the language-system which enables native speakers to construct and understand an indefinitely large number of utterances, including utterances that they have never previously encountered.” (Lyons, 1977)

Challenges of Natural Language Processing

Productivity

Ambiguous

Variability

Diversity

Sparsity

LEVELS:

- Phonological
- Lexical
- Syntactic
- Semantic
- Referential
- Pragmatic

Challenges of Natural Language Processing

Productivity

Ambiguous

Variability

Diversity

Sparsity

Phonological

Words sounds alike:

E.g. (Right, Write), (knows, nose)

Challenges of Natural Language Processing

Productivity

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Variability

Diversity

Sparsity

Lexical

More than one meaning in the language
e.g. Bank (Financial Bank, Edge of a river)

Challenges of Natural Language Processing

Productivity

Ambiguous

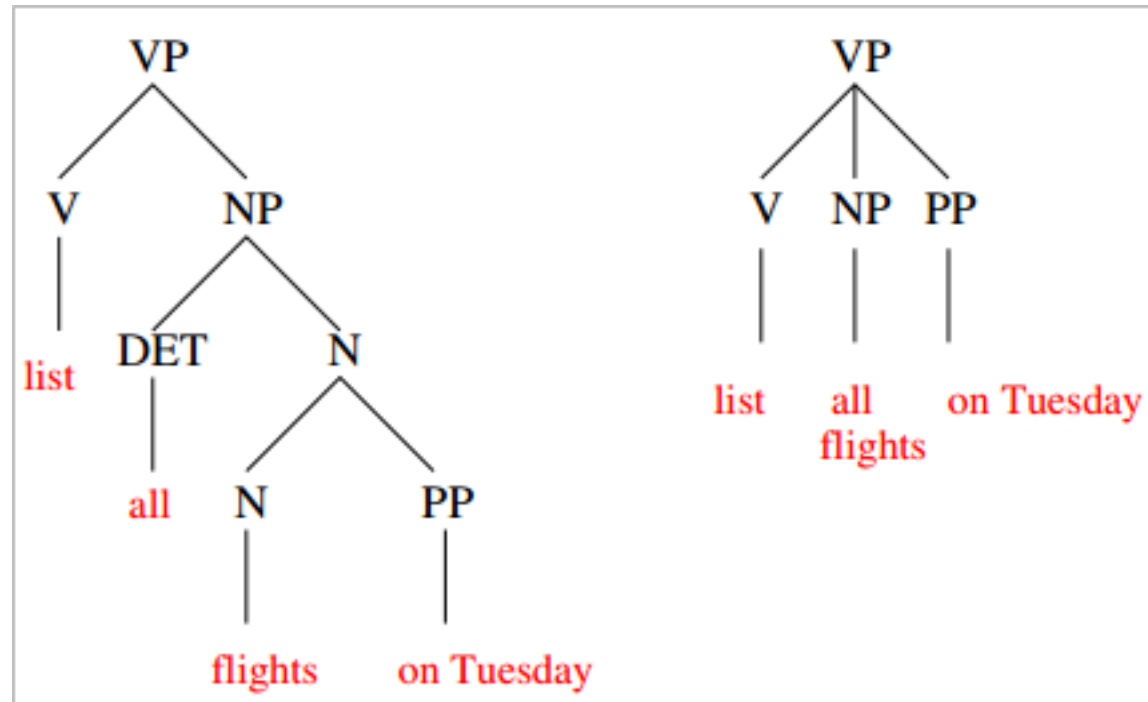
Variability

Diversity

Sparsity

Syntactic

More than one interpretation (at sentence level)



Challenges of Natural Language Processing

Productivity

Ambiguous

Variability

Diversity

Sparsity

Semantic

More than one meaning (given wider context)

Polysemy: e.g. set , arm, head

Head of New-Zealand is a woman

Name Entity: e.g. Michael Jordan:

Michael Jordan is a professor at Berkeley

Object/Color: e.g. cherry

Your cherry coat

Challenges of Natural Language Processing

Productivity

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Diversity

Sparsity

Referential

occurs when a sentence or phrase contains a word or expression that could refer to two or more possible entities, and it's unclear which entity is being referred to.

e.g. "Alice told Jane that her car was parked illegally."

Challenges of Natural Language Processing

Productivity

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Diversity

Sparsity

Pragmatic

occurs when a sentence or phrase is ambiguous due to its context or the way it's used in conversation, rather than because of its structure or the words themselves.

e.g. "I didn't tell John you were late. "

Challenges of Natural Language Processing

Productivity

Ambiguous

Variability

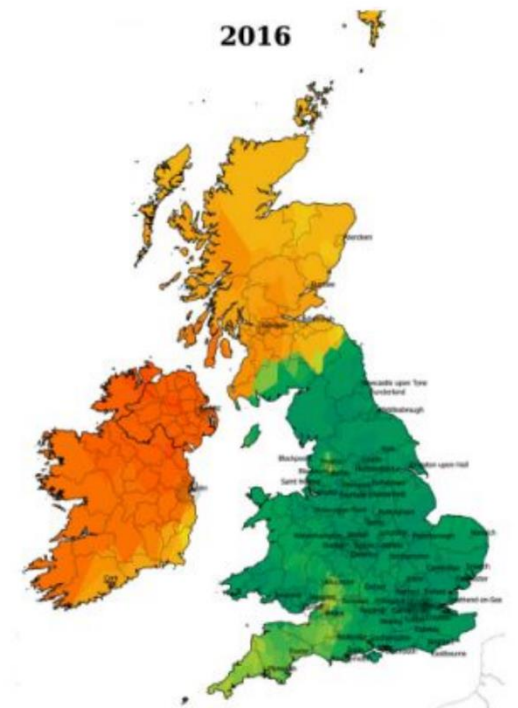
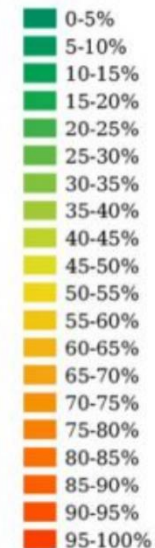
Diversity

Sparsity

LEVELS:

- Phonetic (accent)
- Morphological
- Lexical (spelling)
- Syntactic

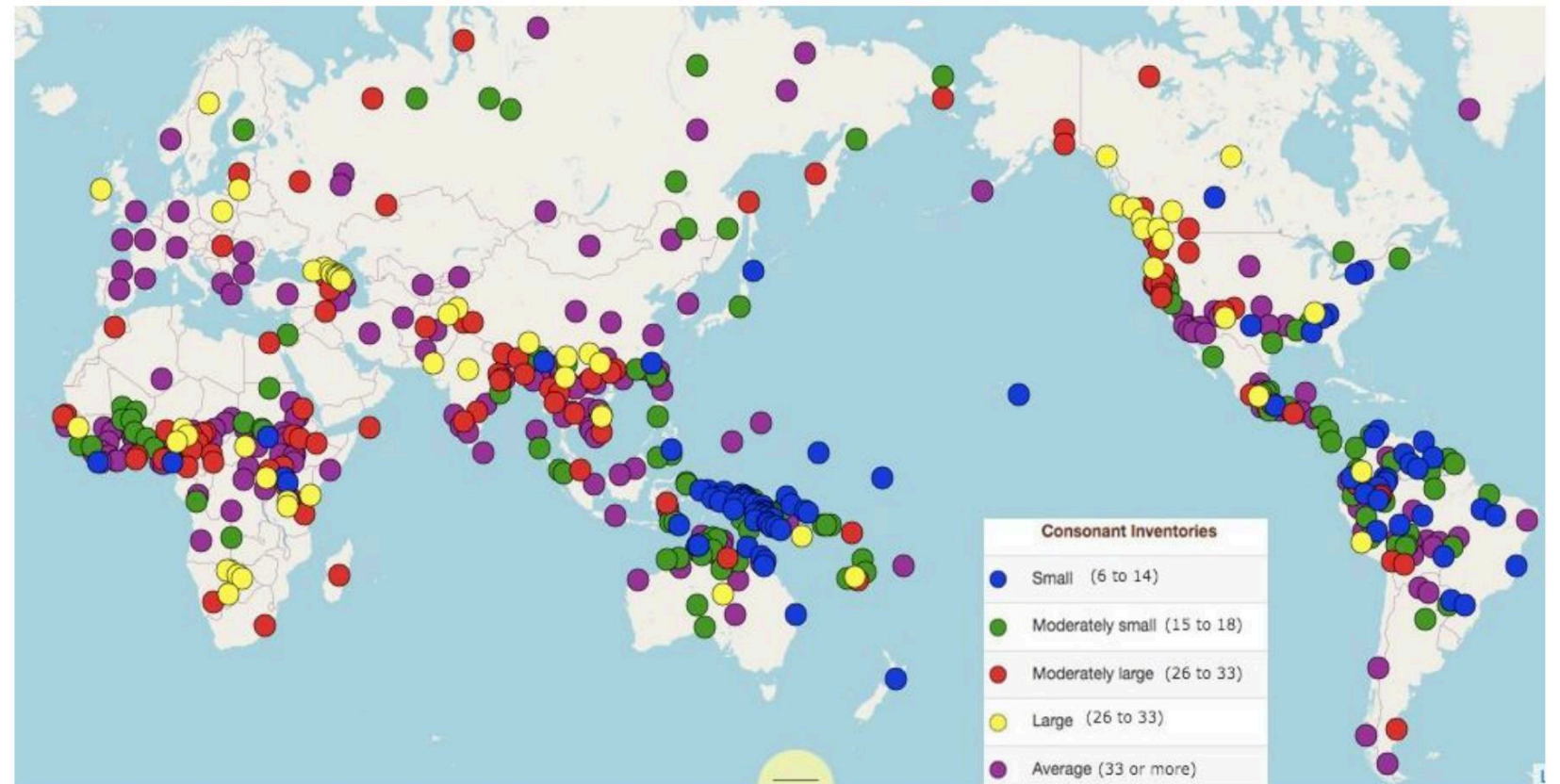
Do you pronounce the
“r” in “arm” ?



Challenges of Natural Language Processing

Productivity
Ambiguous
Variability
Diversity
Sparsity

Phonologic Diversity

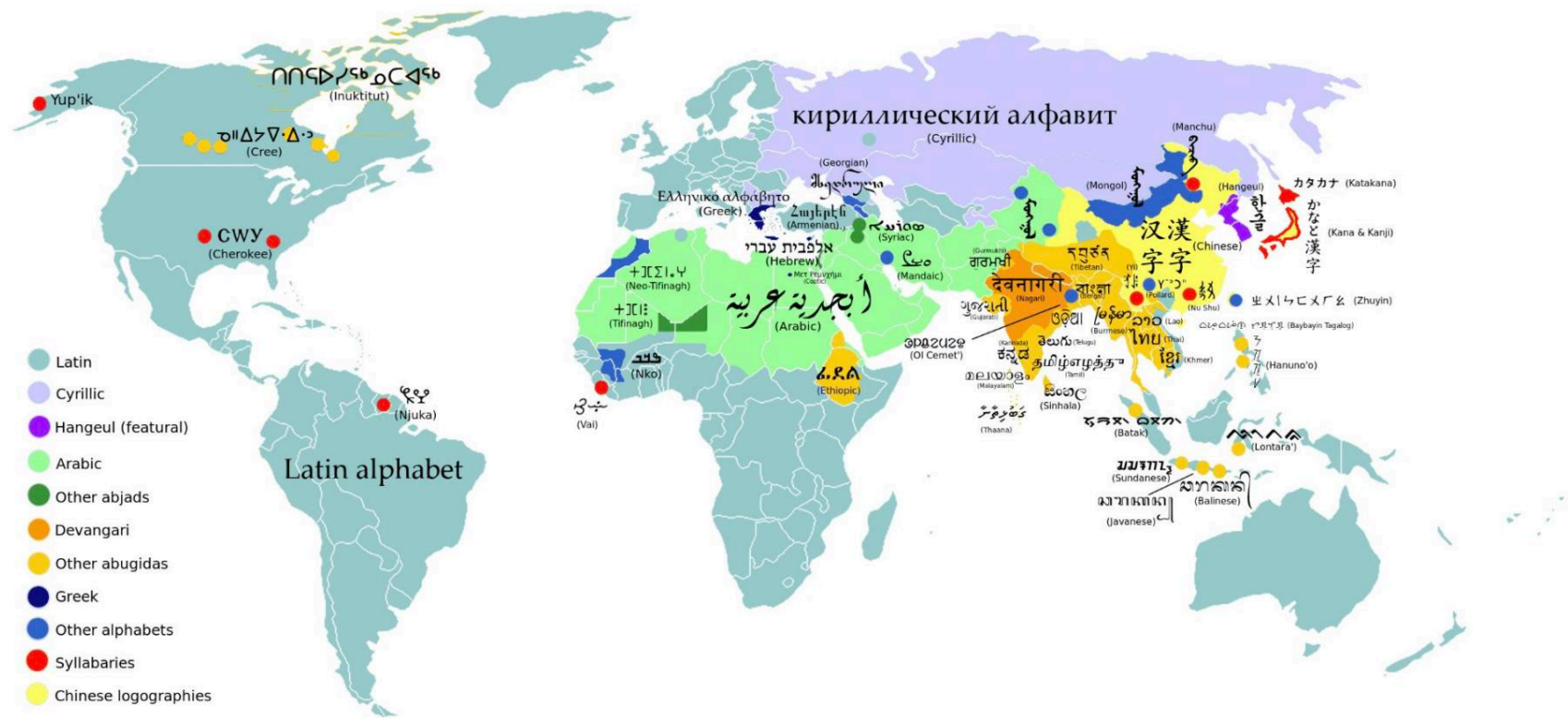


(Dyer et. al 2013)

Challenges of Natural Language Processing

Productivity
Ambiguous
Variability
Diversity
Sparsity

Graphemic Diversity



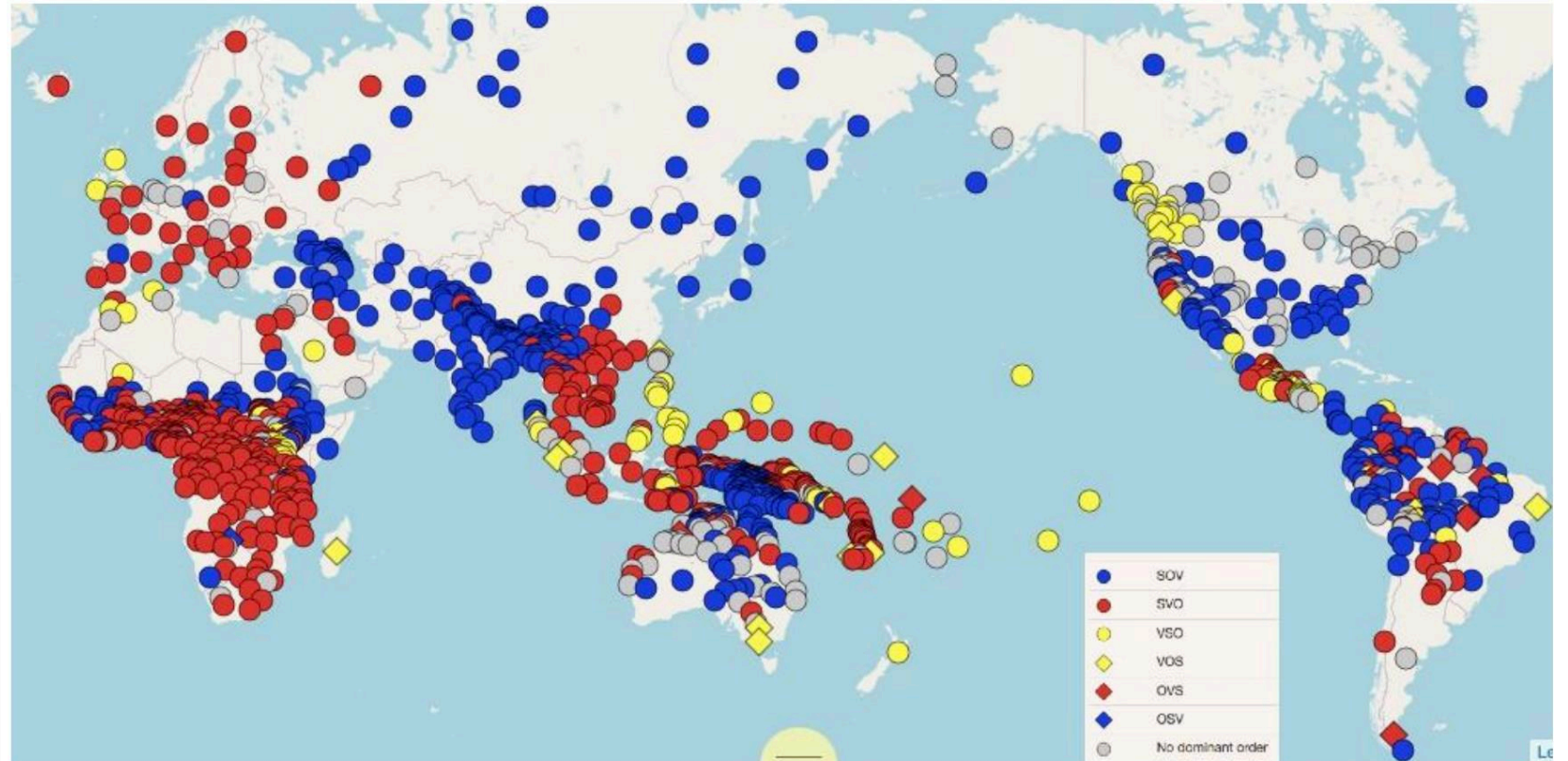
wikipedia

Challenges of Natural Language Processing

Productivity
Ambiguous
Variability
Diversity
Sparsity

Syntactic Diversity

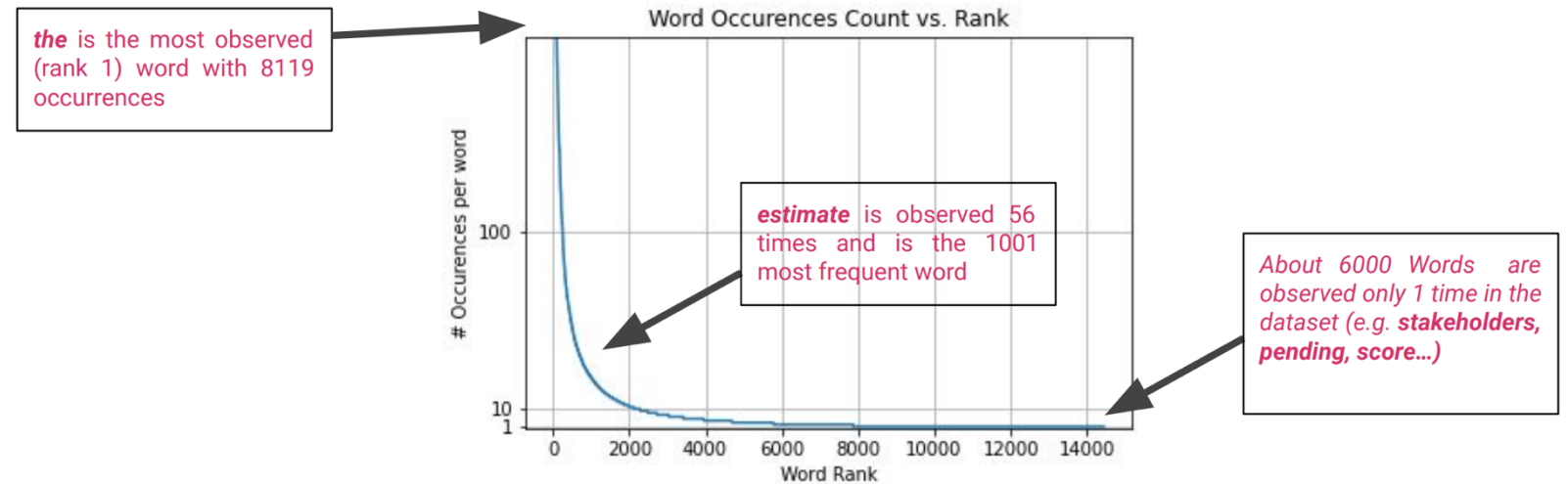
Word order	English equivalent	Proportion of languages	Example languages
SOV	"She him loves."	45%	Japanese, Latin, Tamil
SVO	"She loves him."	42%	English, Mandarin, Russian
VSO	"Loves she him."	9%	Hebrew, Irish, Zapotec
VOS	"Loves him she."	3%	Malagasy, Baure
OVS	"Him loves she."	1%	Apalai?, Hixkaryana?
OSV	"Him she loves."	0%	Warao



Challenges of Natural Language Processing

Productivity
Ambiguous
Variability
Diversity
Sparsity

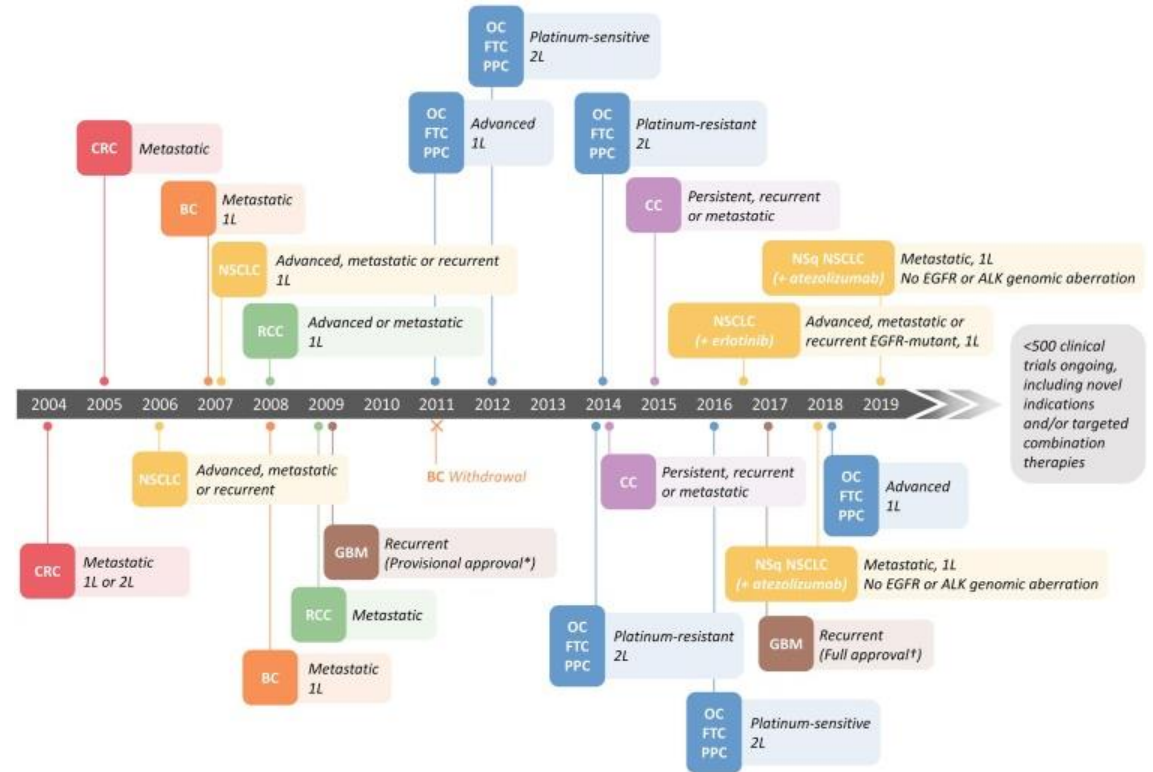
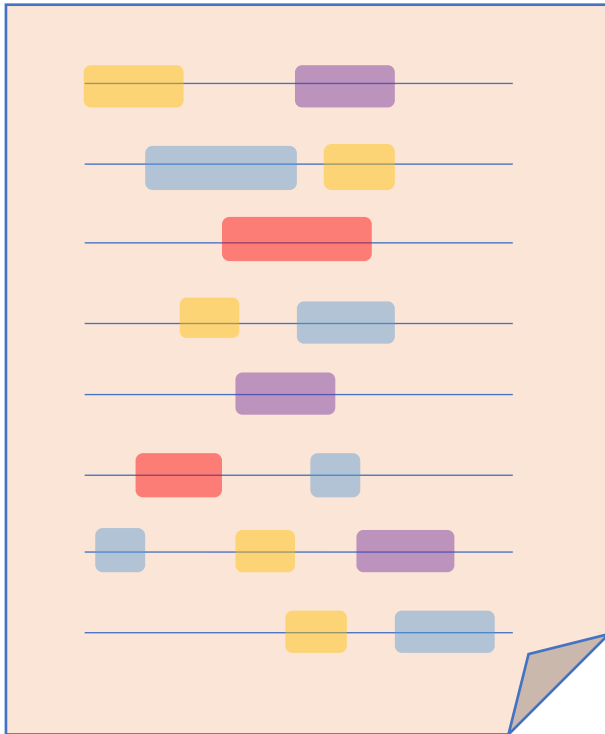
A large number of units are observed in language with very low frequency



Natural Language Processing:

Real-world Use Cases

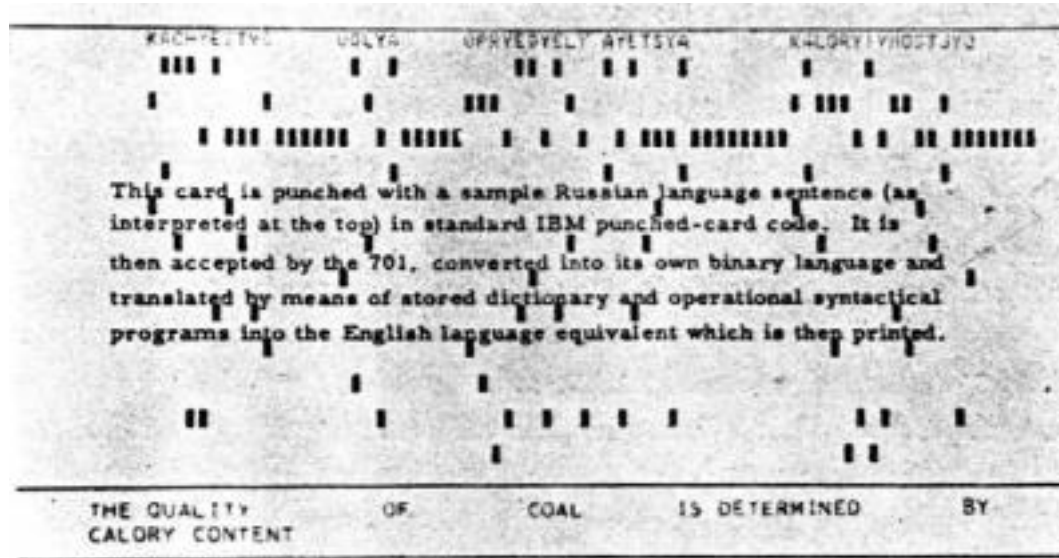
Named Entity Recognition



Natural Language Processing:

Real-world Use Cases

Machine Translation

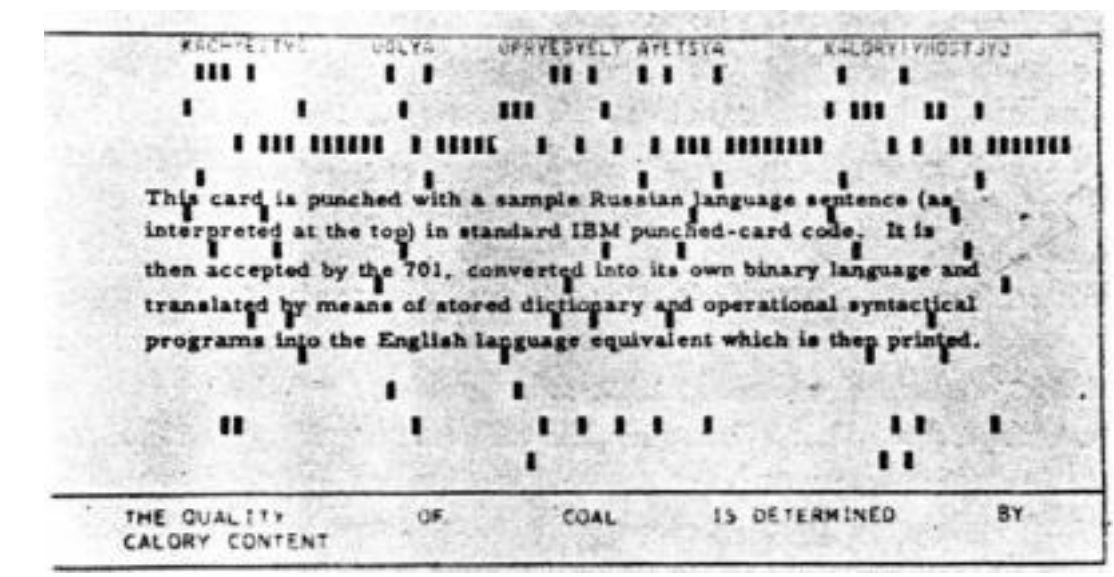


1954: 49 hand-picked Russian sentences to English

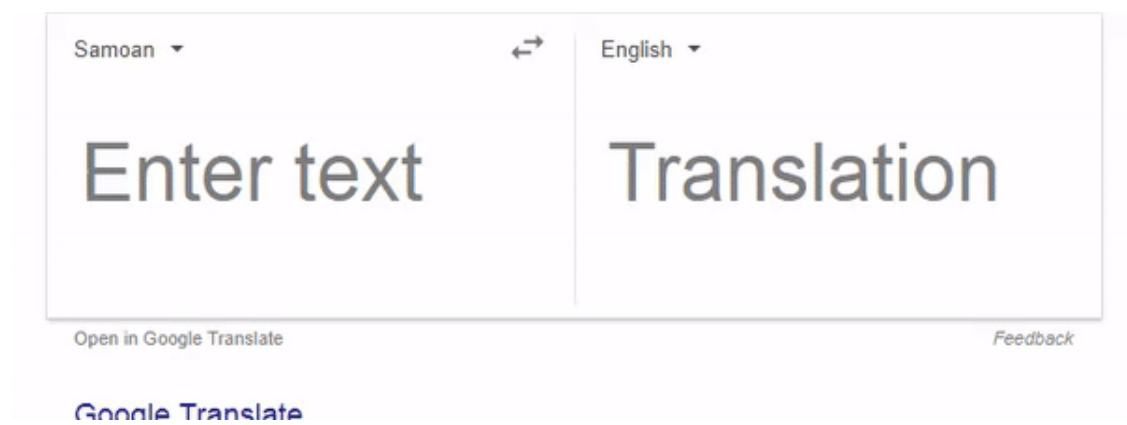
Natural Language Processing:

Real-world Use Cases

Machine Translation



1954: 49 hand-picked Russian sentences to English

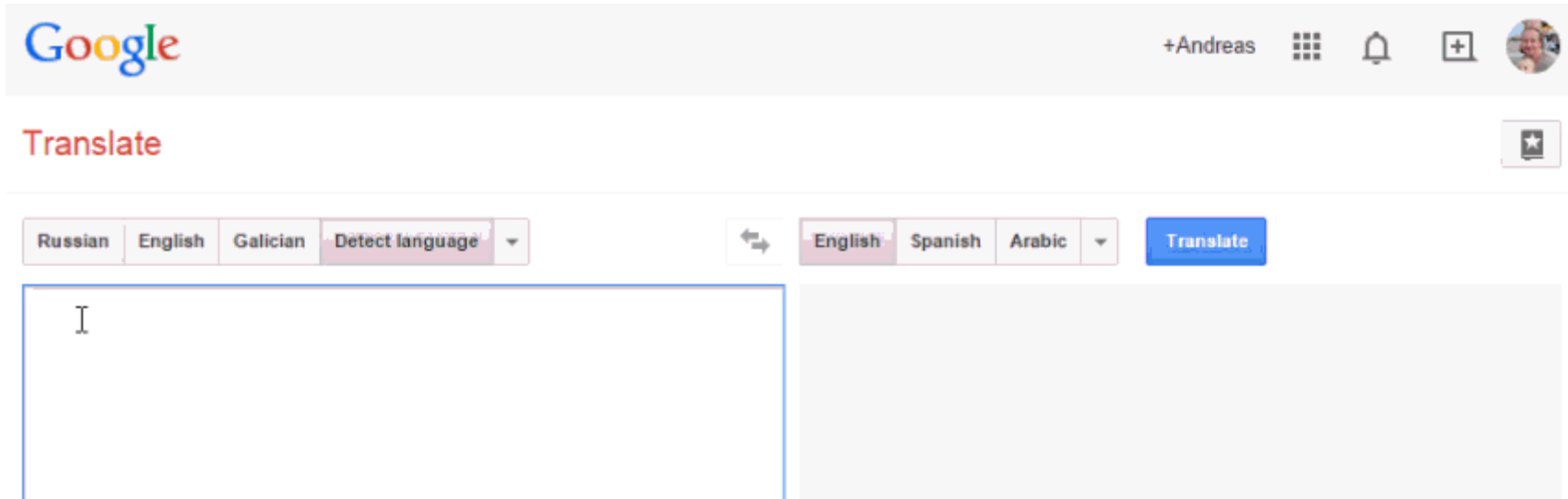


2023

Natural Language Processing:

Real-world Use Cases

Machine Translation



The screenshot displays the Google Translate web interface. At the top, the Google logo is on the left, and user account information (+Andreas) and navigation icons are on the right. Below the header, the word "Translate" is prominently displayed in red. The main interface features a horizontal bar with language selection buttons: "Russian", "English", "Galician", and "Detect language" (which includes a dropdown arrow). To the right of this bar is a swap button (two arrows) and another set of language buttons: "English", "Spanish", and "Arabic" (also with a dropdown arrow). A blue "Translate" button is positioned to the right of the "Arabic" button. Below the language bar, there is a large, empty text input box on the left, containing a single cursor "I". To the right of the input box is a large, empty grey rectangular area for the translated text.

Natural Language Processing:

Real-world Use Cases

Text Summarization

Patient Name: John Doe

Age: 45

Date: 2023-07-12

Chief Complaint: Persistent headache for the past 3 weeks

Medical History: Hypertension, treated with lisinopril; Type 2 Diabetes, managed with metformin

Medications: Lisinopril 10 mg daily, Metformin 500 mg twice daily

Allergies: No known drug allergies

Family History: Father had coronary artery disease, mother has osteoporosis

Social History: Non-smoker, occasional alcohol use

Examination Findings: Blood pressure 140/90, heart rate 78, regular; no neurological deficits noted

Assessment/Plan: Suspected tension-type headache. Advised to monitor blood pressure and glucose levels. Recommended to reduce stress, improve sleep hygiene, and prescribed ibuprofen as needed for pain relief. Follow-up in 4 weeks or earlier if symptoms worsen.

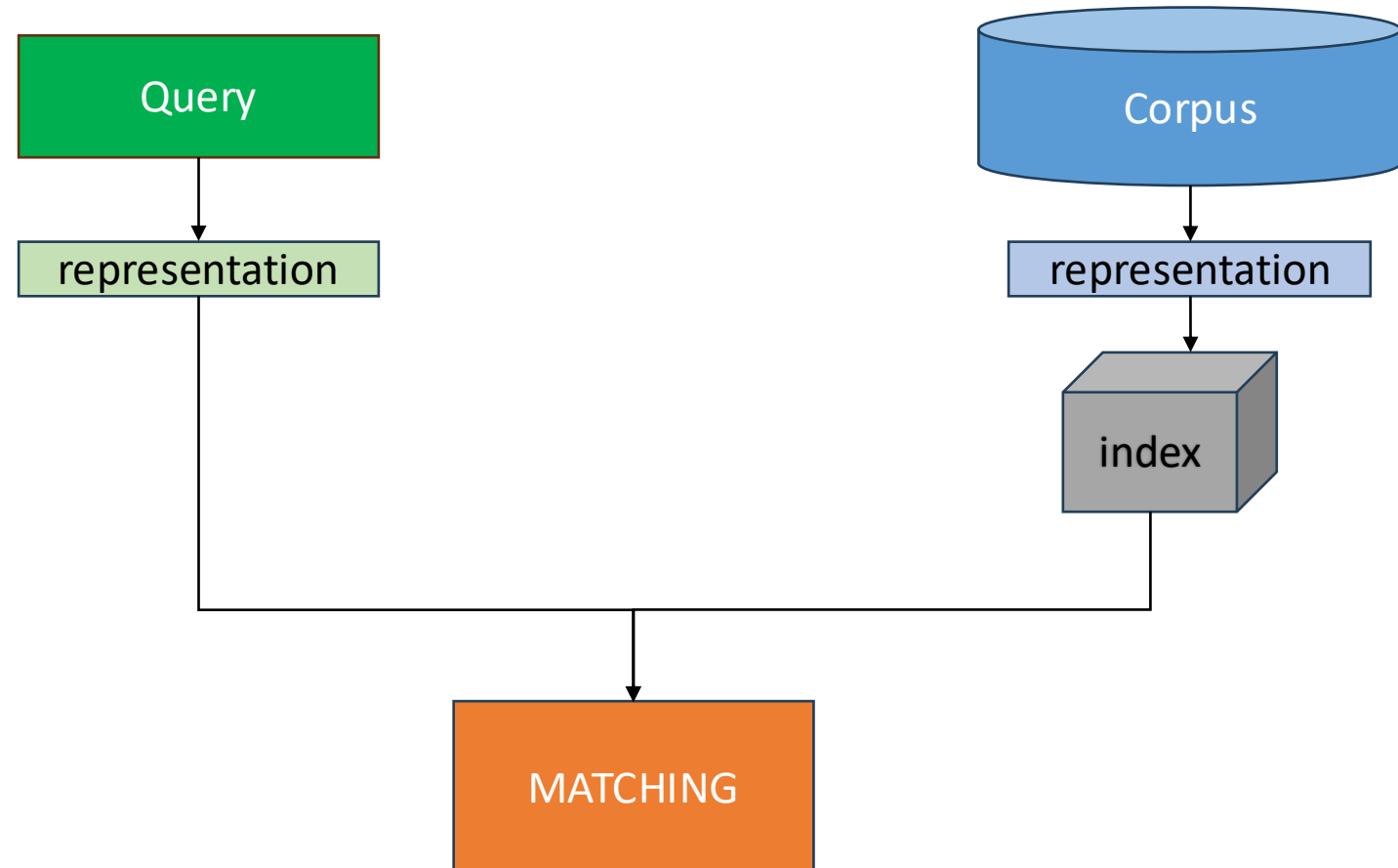


45-year-old male, John Doe, with a history of hypertension and Type 2 Diabetes, presents with a 3-week history of persistent headaches. Vital signs show slightly elevated blood pressure. No neurological deficits. Current medications include Lisinopril and Metformin. Advised to monitor vital signs, reduce stress, improve sleep, and use ibuprofen for pain. Follow-up scheduled in 4 weeks.

Natural Language Processing:

Real-world Use Cases

Information Retrieval



Natural Language Processing:

Real-world Use Cases

Information Retrieval

⚡ AI-Powered GIF Search ⚡

What are you looking for?

Natural Language Processing:

Real-world Use Cases

chatBOT

```
Do you want to log your  
conversation? (Y/[Enter])
```

Natural Language Processing:

Real-world Use Cases

chatBOT

Do you want to log your
conversation? (Y/[Enter])



WHIZARD

Natural Language Processing:

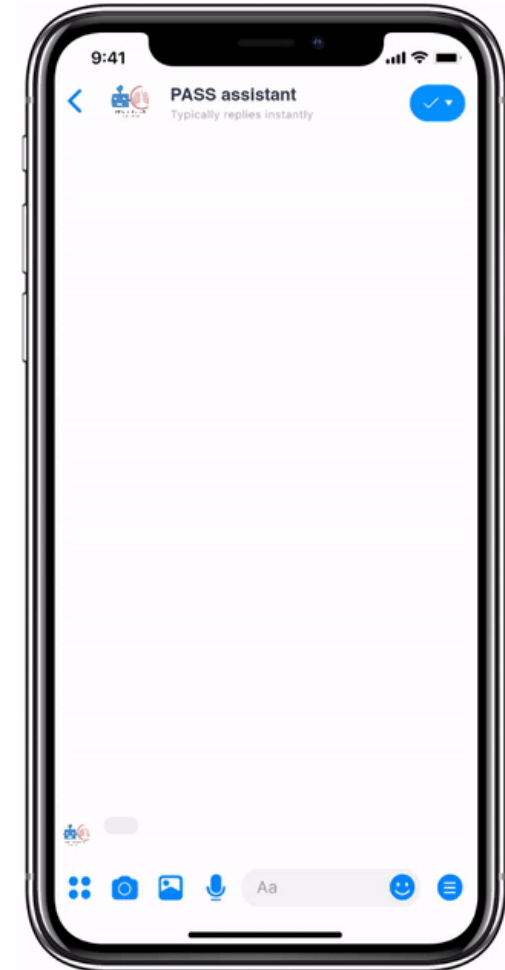
Real-world Use Cases

chatBOT

Do you want to log your
conversation? (Y/[Enter])



WHIZARD

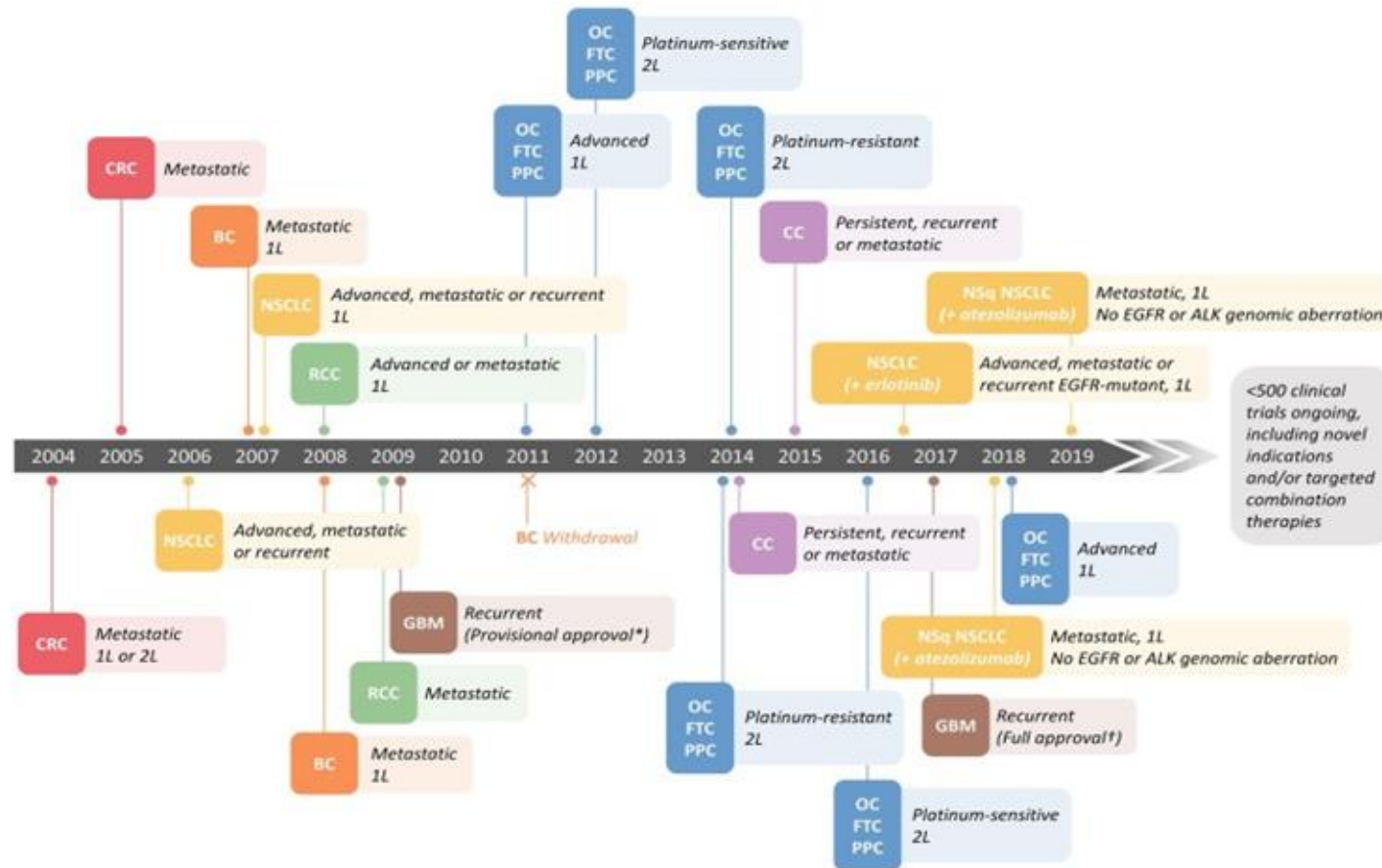


<https://devpost.com/software/pass-pulmonary-diagnosis-screen>

Natural Language Processing:

Clinical Use Case

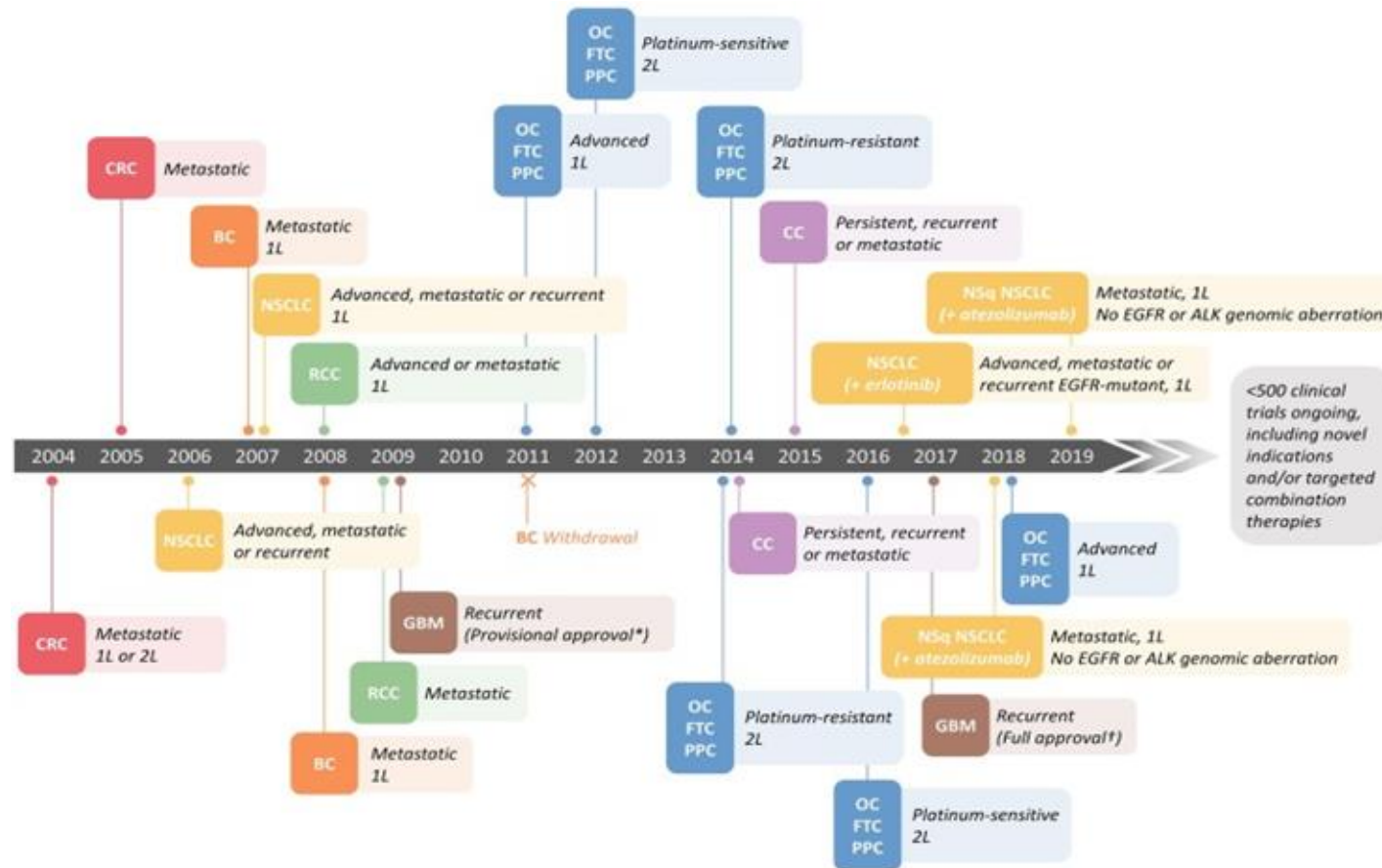
Objective: Automate structuring of patient health records



Natural Language Processing:

Clinical Use Case

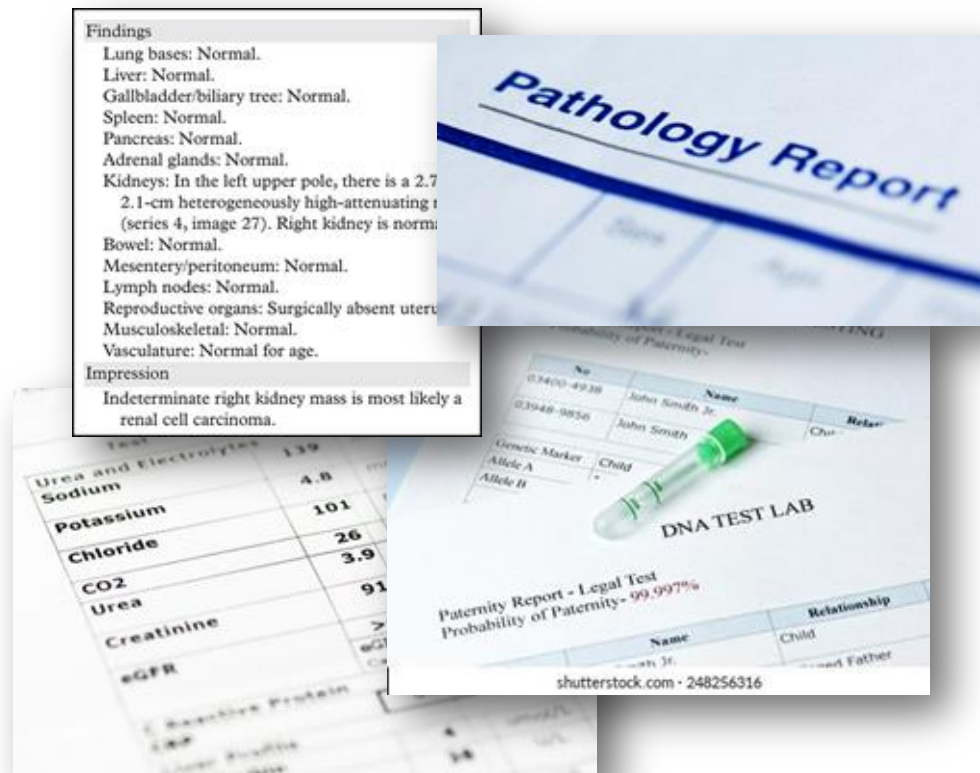
Objective: Automate structuring of patient health records to achieve human-level performance



Natural Language Processing:

Clinical Use Case

Input



PDF

Fax copies

Document images

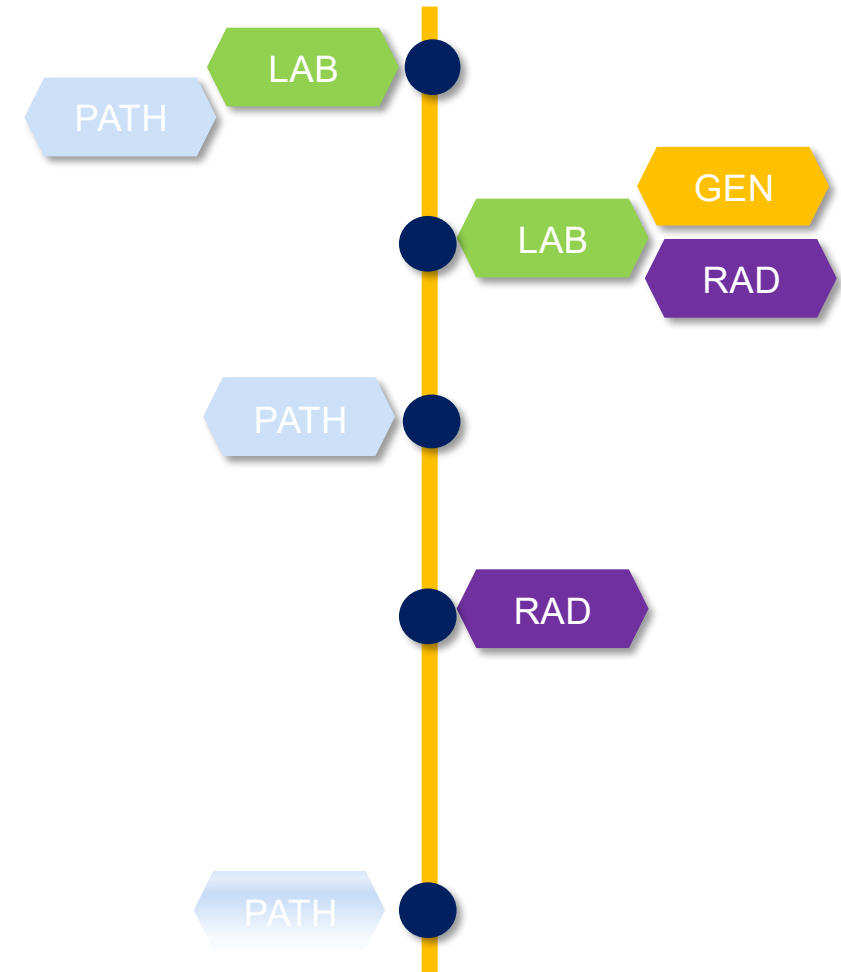
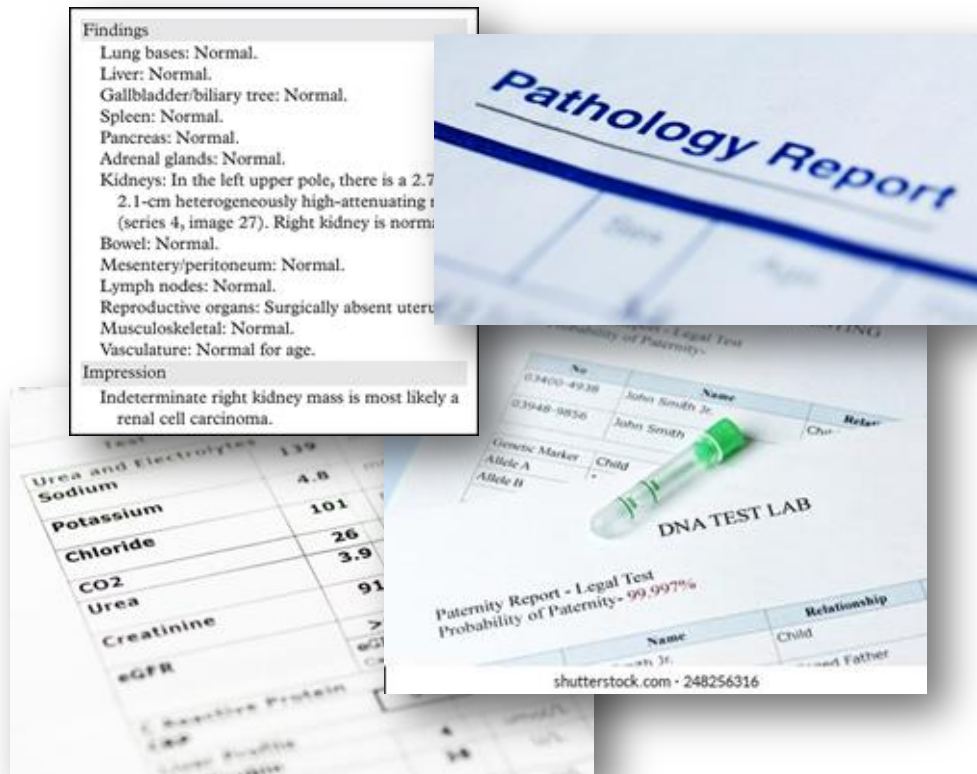
HL7 messages

...

Natural Language Processing:

Clinical Use Case

Output



Natural Language Processing:

Clinical Use Case

Challenges

- **Capture format:** handwritten/digital, templated/free-style, etc.
- **Quality:** digital pdf→ scanned
- **Data format:** free-text, tables, key-value pairs, etc.
- **Document variety:** Rad reports, Path reports, Lab results, etc.
- **Technical language:** Radiologist 1 vs. Radiologist 2
- **Written language:** English, French, Spanish, etc.

The image shows two overlapping forms titled "OCCUPATIONAL THERAPY ASSESSMENT AND RE-ASSESSMENT". The top form is filled out with handwritten notes. It includes sections for "AREAS OF ASSESSMENT", "VARIANCE", "GOALS", "TREATMENT PLAN", and "THERAPIST SIGNATURE". The bottom form is partially visible and also contains handwritten notes. The forms are from the City of Hope National Medical Center.

Natural Language Processing:

Clinical Use Case

Challenges

- Data Size
- Data Variety
- Data Labels

